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✉ For correspondence:

sijia.zhao@psy.ox.ac.uk

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Dynamic updating of spatial working memory across eye movements: a computational investigation of transsaccadic integration

Sijia Zhao^{1,‡}✉, Thomas Parr^{2,‡}, Rob Udale¹, Verena Klar¹, Gabriel David Jones^{2,3}, Anna Scholcz^{1,2}, Sofia Toniolo^{2,4}, Sanjay G Manohar^{1,2,4}, Masud Husain^{1,2,4}

¹Department of Experimental Psychology, University of Oxford, Oxford, United Kingdom • ²Nuffield Department of Clinical Neurosciences, University of Oxford, Oxford, United Kingdom • ³Nuffield Department of Women's and Reproductive Health, University of Oxford, Oxford, United Kingdom • ⁴Cognitive Disorders Clinic, John Radcliffe hospital, University of Oxford, Oxford, United Kingdom

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This study makes an **important** contribution by revealing how saccades selectively disrupt spatial working memory while sparing other object features, and by demonstrating how this mechanism is altered in aging and neurodegeneration. The findings are supported by **convincing** evidence derived from well-controlled eye-tracking experiments and systematic generative model comparisons. Together, the work provides a computationally grounded framework that is of importance for understanding trans-saccadic memory and its clinical relevance.

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Abstract

The brain continuously integrates rapidly changing visual input across eye movements to maintain stable perception, yet the precise mechanisms underpinning dynamic working memory and how these break down in brain diseases remain unclear. We developed a novel eye-tracking paradigm and computational models to investigate how spatial and colour information are updated across saccades. Our findings reveal that saccades selectively impair spatial but not colour memory. Computational modelling identified that spatial representations are maintained in a dual eye-centred frame of reference which is actively updated by a noisy memory of saccades but is vulnerable to interference. Using this model, we found that specific mechanistic failures in initial encoding and memory decay, rather than the saccadic updating process itself, account for spatial working memory deficits in Alzheimer's and Parkinson's disease. These results provide a mechanistic understanding of how dynamic spatial memory operates in health and its disruption in neurodegenerative disorders.

Introduction

Natural vision is profoundly dynamic, relying upon sequences of saccades. Everyday tasks, such as making a cup of tea,^{1,2} arranging the layout of objects,^{3,4} or copying a simple line drawing^{5–7}, require us to precisely track spatial locations—which appear at different retinotopic locations with each fixation—across saccades.⁸ This requires a dynamic updating process, termed “remapping”, that involves updating our internal representation of object locations to maintain a stable world-centred frame of reference.^{9–12} One key hypothesis is that we use working memory across visual fixations to update perception dynamically.^{13–15} However, despite the apparent stability of our visual world, it has proven difficult to separate the distinct contributions of retinotopic memory

from transsaccadic memory. To precisely characterise the interplay of factors contributing to dynamic spatial working memory—particularly amidst the complexities of multiple fixations as well as the potential for memory decay and interference—and to move beyond purely descriptive observations, fundamental questions regarding the underlying mechanisms and sources of potential errors need to be addressed.

Central to this enquiry is the nature of the coordinate system used for the brain's internal spatial representation. Does the brain maintain a single, world-centred (allocentric) map, or does it rely on a dynamic, eye-centred (retinotopic) representation?^{11,13,16,17} In the latter system, retinotopic memory preserves spatial information within a fixation, whereas transsaccadic memory describes the active process of updating these representations across eye movements to achieve spatiotopic stability—the perception of a stable world despite eye movements.^{11,17–19} If spatial stability is indeed reconstructed through such remapping, the mechanism remains unresolved: do we retain memories of absolute fixation locations, or do we reconstruct these positions from noisy memories of the intervening saccade vectors? We can test these hypotheses by analysing when and where memory errors occur. Assuming that memory precision declines over time,²⁰ the resulting error distributions should reveal the specific variables that are represented and updated across each saccade.

The clinical relevance of these spatial mechanisms is underscored by significant disruptions to visuospatial processing and constructional apraxia—a deficit in copying and drawing figures—observed in neurodegenerative conditions such as Alzheimer's disease (AD) and Parkinson's disease (PD).^{21,22} This raises a crucial question: do clinical impairments in complex visuomotor tasks stem from specific failures in transsaccadic remapping? If so, the computational parameters that define normal spatial updating should also provide a mechanistic account of these clinical deficits, differentiating them from general age-related decline.

To address these questions and move beyond purely descriptive observations towards mechanistic models, we developed a novel eye-tracking paradigm—the Location and Colour Updating across Saccades (LOCUS) paradigm (Figure 1 [↗](#)). This paradigm allows us to directly compare retinotopic and transsaccadic working memory for both spatial location and colour. Combined with computational modelling, our approach allowed us to precisely characterize the interplay of factors contributing to dynamic spatial working memory, isolate the impact of saccades, and test specific, competing hypotheses regarding transsaccadic memory. Including young and older healthy adults as well as patients with Parkinson's and Alzheimer's disease, our findings reveal that saccades selectively disrupt spatial but not colour memory.

Across all healthy and patient groups, the winning computational model indicated that spatial representations are maintained in a dual (retinotopic) reference frame, updated based on a noisy memory of the saccade vector, and susceptible to interference from other items in memory. Importantly, we identified specific model parameters—related to angular encoding, memory decay, and interference—that differentiated the patient groups from healthy controls, providing a mechanistic account of their visuospatial deficits. Finally, by linking these mechanistic parameters to a standard clinical measure of constructional ability (the Rey-Osterrieth Complex Figure task), we demonstrate that transsaccadic updating represents a core computational phenotype underpinning real-world visuospatial construction in both health and neurodegeneration.

Results

Saccades disrupt spatial memory but spare colour

In the LOCUS (Location and Colour Updating across Saccades) paradigm (Figure 1 [↗](#)), participants memorised the colour and location of two sequentially presented squares. Subsequently, they were cued to recall the colour and location of one of these squares, i.e. either the first or the second. Colour recall involved clicking on a colour wheel to indicate the remembered colour using the mouse. Location recall required dragging a colour-matched square to the remembered position on the screen.



Figure 1. Location and Colour Updating across Saccades (LOCUS) Task.

Participants were asked to fixate a white cross, wherever it appeared. They had to remember the colour and location of a sequence of two briefly presented coloured squares (Item 1 and 2), each appearing within a white square frame. They then fixated a colour wheel wherever it appeared on the screen, which served as the target for the second instructed saccade (i.e., a movement from the second fixation cross to the colour wheel location). This cued recall of a specific square (Item 1 or Item 2 labelled within the colour wheel). Participants selected the remembered colour on the colour wheel which led to a square of that colour appearing on the screen. They then dragged this square to its remembered location on the screen. Saccadic demands were manipulated by varying the locations of the second frame and the colour wheel, resulting in four conditions in their reliance on retinotopic versus transsaccadic memory: **(1)** No-Saccade condition providing a baseline measure of within-fixation precision as no eye movements were required. **(2)** Saccade After Item 1; **(3)** Saccade After Item 2; **(4)** Saccades after both items (Two Saccades condition). In all conditions requiring eye movements, saccade vectors were constrained to a minimum amplitude of 8.5° (degrees of visual angle). While the No-Saccade condition isolates retinotopic working memory, conditions (2) to (4) collectively quantify the impact of varying saccadic demands and timings on the maintenance of spatial information, thereby assessing the efficacy of the transsaccadic updating process.

By manipulating the spatial location of the squares and also that of the colour wheel, we controlled saccadic demands on every trial, creating four conditions with equal probability: no-saccade, saccade after Item 1, saccade after Item 2 (to the colour wheel), or a saccade after both items. The no-saccade trials provided a measure of retinotopic working memory, while the two-saccade trials assessed transsaccadic working memory.

To mitigate potential confounds, we monitored eye position throughout the experiment. Eye-tracking analysis confirmed high compliance in healthy adults, who followed instructions on the vast majority of trials (Younger Adults: $97.2 \pm 5.2\%$; Older Adults: $91.3 \pm 20.4\%$). The mean difference between these groups was negligible, representing just 1.25 trials per condition, and was not statistically significant ($t(80) = 0.16$, $p = 1.000$; see more in *Methods – Eyetracking data analysis*). Non-compliant trials were excluded from all further analyses.

This design allowed us to isolate and quantify the unique impact of saccades on spatial memory, enabling us to test competing hypotheses regarding spatial representation. If spatial memory were solely underpinned by an allocentric mechanism, precision should remain comparable across all conditions as the representation would be world-centred and unaffected by eye movements. Thus, performance in the no-saccade condition should be comparable to the two-saccade condition. Conversely, if spatial memory relies on a retinotopic representation requiring active updating across eye movements, the two-saccade condition was anticipated to be the most challenging due to cumulative decay in the memory traces used for stimulus reconstruction after each saccade.²³ Critically, we hypothesised that this saccade cost would be specific to the spatial domain; while location requires active remapping via noisy oculomotor signals, non-spatial features like colour are not inherently tied to coordinate transformations and should therefore remain stable (see more in Discussions below).

Meanwhile, the no-saccade condition was expected to yield the most accurate localisation, relying solely on retinotopic information (retinotopic working memory). These predictions were confirmed in young healthy adults ($N = 21$, mean age = 24.1 years, ranged between 19 and 34; Figure 2A and Figure 2B). A repeated measures ANOVA revealed a significant main effect of saccades on location memory ($F(3, 20) = 51.52$, $p < 0.001$, partial $\eta^2 = 0.72$), with Bayesian analysis providing decisive evidence for the inclusion of the saccade factor ($BF_{\text{incl}} = 3.52 \times 10^{13}$, $P(\text{incl} | \text{data}) = 1.00$). In contrast, colour memory remained remarkably stable across all saccade conditions ($F(3, 20) = 0.57$, $p = 0.64$, partial $\eta^2 = 0.03$). This null effect was supported by Bayesian analysis, which provided moderate evidence in favour of the null hypothesis ($BF_{01} = 8.46$, $P(\text{excl} | \text{data}) = 0.89$), indicating that the data were more than eight times more likely under the null model than a model including saccade-related impairment.

This “saccade cost”—the loss of memory precision following an eye movement—indicates that spatial representations require active updating across saccades rather than being maintained in a static, world-centred reference frame.

Critically, our comparison between spatial and colour memory does not rely on the absolute magnitude of errors, which are measured in different units (degrees of visual angle vs. radians). Instead, we assessed the relative impact of the same saccadic demand on each feature within the same trial. While location recall showed a robust saccade cost, colour recall remained statistically unchanged. To ensure this null effect was not due to a lack of measurement sensitivity, we examined the recency effect; recall performance for the second item was predicted to be better than for the first stimulus in each condition.^{24,25} As expected, colour memory for Item 2 was significantly more accurate than for Item 1 ($F(1,20) = 6.52$, $p = 0.02$, partial $\eta^2 = 0.25$), demonstrating that the task was sufficiently sensitive to detect standard working memory fluctuations despite the absence of a saccade-induced deficit.

Regarding the effect of item order on location error, saccades affected both items similarly, with no interaction between saccades and item ($F(2.6, 52.7) = 0.64$, $p = 0.57$, partial $\eta^2 = 0.03$). However, as expected, spatial memory for Item 1 was significantly less accurate than for Item 2 ($F(1,20) = 12.5$, $p = 0.002$, partial $\eta^2 = 0.38$). Crucially, even a single saccade after Item 2 significantly reduced location memory accuracy compared to the no-saccade condition ($t = -7.64$, $p(\text{bonf}) < 0.001$).

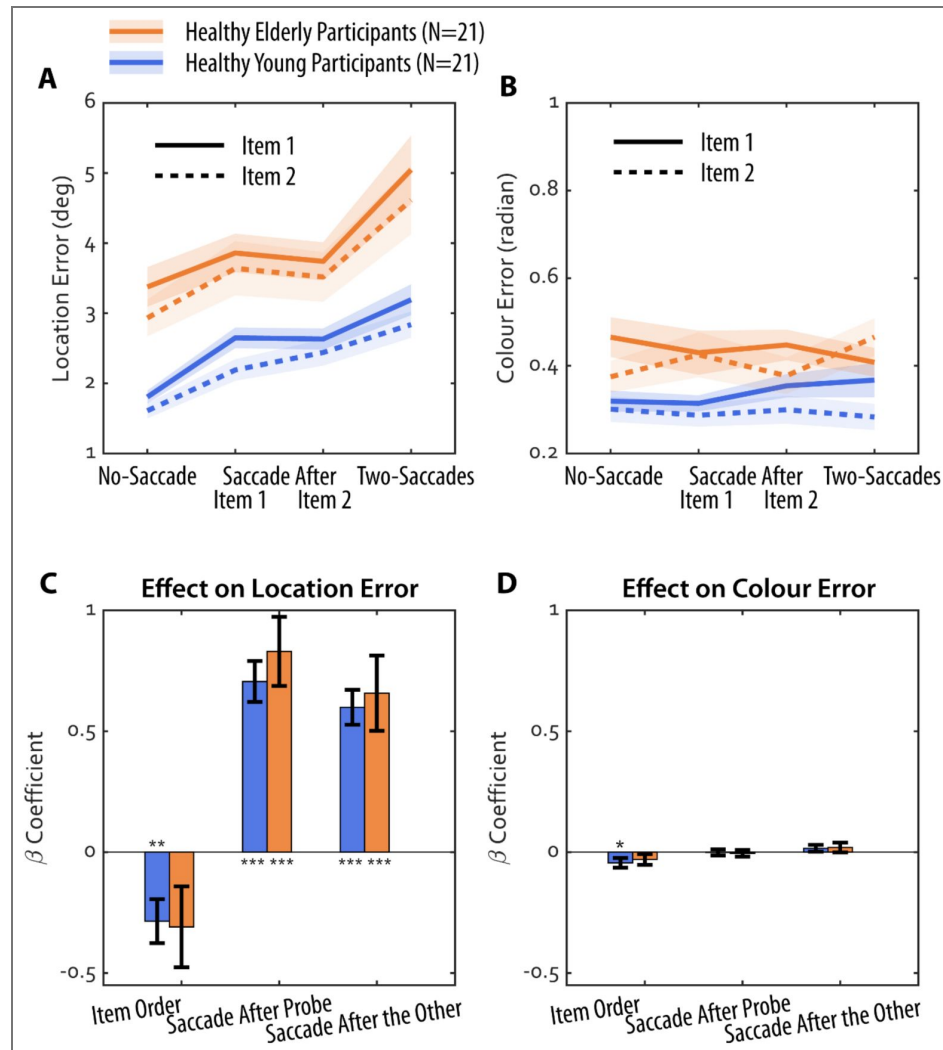


Figure 2. Saccades lead to increased recall error for location but not colour.

(A) Mean location error for each saccade condition, and for each probed item (Item 1: solid lines; Item 2: dashed lines). **(B)** Colour error for the same conditions. Shaded error represents ± 1 standard error from mean. **(C)** Beta coefficients from a multiple linear regression model predicting location error, illustrating the effect of item order (1 or 2), saccade after the probed item, and saccade after the other item, for young (blue) and elderly (orange) participants. Error bars represent ± 1 standard errors. **(D)** The same model applied to colour errors, revealing a recency effect but no impact of saccade demands. ** means significantly different from zero $p < 0.01$, *** means significantly different from zero $p < 0.001$.

Moreover, larger saccades were associated with increased spatial memory disruption (see Supplementary Materials for “*Larger saccades disrupt memory more*” and Supplementary Figure 1). A multiple linear regression predicting location error showed that only the vertical distance between stimulus frames significantly predicted error ($t(41) = 5.73, p < 0.0001$), with marginal significance for horizontal saccade direction ($t(41) = -1.94, p = 0.059$). This suggests that greater displacement, particularly vertical, exacerbates spatial memory impairment, irrespective of saccade direction. However, there was no evidence that crossing saccades (e.g., rightward then leftward) or specific horizontal/vertical saccade directions differentially affected spatial memory precision, as revealed by repeated measures ANOVAs ($F(1.3, 49.8) = 0.04, p = 0.89$, partial $\eta^2 = 0.001$ for horizontal crossing; $F(1.7, 62.8) = 0.29, p = 0.71$, partial $\eta^2 = 0.008$ for vertical crossing).

Further analysis examined whether individual differences in baseline memory precision (no-saccade condition) predicted resilience to saccadic disruption. Crucially, individual saccade costs (defined as the precision loss relative to baseline) did not correlate with baseline precision ($\rho = 0.20, p = 0.20$). This suggests that the noise introduced by transsaccadic remapping acts as an independent, additive source of variance that is not modulated by an individual’s underlying memory capacity. These findings imply a functional dissociation between the mechanisms responsible for maintaining a representation and those involved in its coordinate transformation.

Saccadic interference with spatial memory is age-independent

To investigate the impact of aging, we replicated the experiment in a group of healthy adults above 60 (Elderly Healthy Adults; $N = 21$; mean age = 72.4 years, ranged between 60 and 80). A three-way ANOVA (4 saccade conditions $\times$ 2 items $\times$ 2 age groups) revealed a main effect of age, with older participants demonstrating lower overall accuracy compared to their younger counterparts (*Figure 2A*, $F(1.48, 20) = 8.38, p = 0.003$, partial $\eta^2 = 0.30$, Greenhouse-Geisser sphericity correction applied), with Bayesian analysis providing decisive evidence for the inclusion of the saccade factor ($BF_{\text{incl}} = 251.4, P(\text{incl} | \text{data}) = 1.00$). This age-related decline was evident in both transsaccadic working memory (Two-Saccades Location Error, $\rho = 0.57, p < 0.001$, Fisher’s $z = 0.64$; partial $\rho = 0.55, p < 0.001, z = 0.62$ after controlling gender and ACE total score) and retinotopic working memory (No-Saccade Location Error, $\rho = 0.70, p < 0.001$, Fisher’s $z = 0.86$; partial $\rho = 0.69, p < 0.001, z = 0.85$).

In contrast, colour memory remained unaffected by saccade demands ($F(3, 20) = 0.57, p = 0.65$, partial $\eta^2 = 0.03$), again supported by the Bayesian analysis: $BF_{01} = 8.68, P(\text{excl} | \text{data}) = 0.90$.

These findings suggest that while both retinotopic and transsaccadic working memory decline with age, the relative impact of saccades on spatial memory remains stable across age groups. This was also supported by the lack of significant interaction between age group, saccade condition, and item type (all $p > 0.23$), suggesting that the detrimental effect of saccades on spatial memory is constant across both young and older adults. This observation was further supported by post-hoc tests, which confirmed the *same* pattern of location errors across saccade conditions in both groups: the no-saccade condition exhibited the least location error ($t(20) \geq 5.94, p(\text{bonf}) < 0.001$), followed by the one-saccade conditions (no difference between them: $t(20) = 0.02, p(\text{bonf}) = 1.00$, Cohen’s $d = 0.002$), and with the highest error on the two-saccade condition ($t(20) \geq 3.53, p(\text{bonf}) \leq 0.006$).

A multiple linear regression model, incorporating data from both age groups further corroborated these findings. Both earlier item presentation and the presence of saccades after either item predicted increased location error (*Figure 2C*) but not colour error (*Figure 2D*). There was no difference in the beta coefficients between two age groups.

In summary, while healthy aging was associated with a general decline in spatial memory accuracy, the specific disruptive effect of saccades on spatial – but not colour – memory remained constant across age groups. Thus, the mechanisms underlying transsaccadic remapping and its impact on spatial working memory are generally preserved in healthy aging, despite an overall reduction in memory performance.

Computational model comparison reveals the mechanisms of transsaccadic memory

To dissect the specific cognitive processes underlying the observed saccade cost on spatial memory, we turned to computational modelling with a suite of seven computational models designed to test specific, competing hypotheses about how the brain represents and updates visual information across saccades (Table 1 [↗](#)). The models are organised hierarchically, starting from a simple, single-representation model and progressively incorporating more complex mechanisms. This allows us to systematically identify which mechanisms are necessary to account for the observed behaviour. The model, data, and analysis scripts are all available in the open dataset: <https://osf.io/95ecp/> [↗](#).

Our modelling framework addresses three core conceptual distinctions:

1. **Reference frame.** Whether spatial memory is maintained in a single, world-centred (allocentric) map (Figure 3A [↗](#)), or relies on a dual, fixation-centred (retinotopic) reference frame (Figure 3B [↗](#)).
2. **Distractor interference.** Whether memory is susceptible to interference from other items held in memory (Figure 3C [↗](#)).
3. **Remapping source.** Within a dual-frame system, whether error arises from a passive decay of the remembered fixation point (Figure 3D [↗](#)), or from an active but noisy reconstruction based on the saccade vector itself (Figure 3E [↗](#)).

All models assume that memory for location decays over time, formalised as a diffusion (Brownian motion) process, where the precision of the remembered location decreases with increasing retention interval. The key differences between the models lie in what is decaying and when (see Methods for mathematical details and detailed variance functions).

These conceptual distinctions give rise to a family of seven specific computational models (Table 1 [↗](#)):

1. **Allocentric model:** This model assumes the brain stores locations in a single, stable, world-centred (allocentric) reference frame. In this model, memory error arises solely from a time-dependent decay of the stored coordinates, independent of any eye movements.
2. **Dual model:** This foundational model introduces the core hypothesis of a dual-reference frame. It posits that locations are initially encoded in eye-centred (retinotopic) coordinates (**radial** distance and **angular** direction from fixation), which are then subject to both initial encoding errors and time-dependent decay. This forms the basis for more complex models that explicitly distinguish between decay in the reference frame's centre and coordinates within that frame.
3. **Dual (fixation decay) model:** Building on the dual-frame hypothesis, this model proposes that to remap a location after a saccade, the brain uses its memory of the *original fixation point*. This memory trace of the eye's position itself decays over time, introducing a specific source of error into the remapped location. Of note, the time at which the memory of the fixation location starts to decay is the time of the first saccade following the disappearance of the stimulus. This means that, depending upon the experimental condition, the fixation memory starts to decay *after* the retinotopic memories start to decay.
4. **Dual (saccade update) model:** This model offers an alternative updating mechanism. Instead of relying on a memory of the absolute fixation location, it proposes that the brain uses a memory of the *saccade vector* itself (the direction and amplitude of the eye movement). The original location is then reconstructed based on this imperfect, noisy memory of the saccade, providing a different source of remapping error. Here, each saccade between the fixation at which the stimulus was shown and the response time injects an additional encoding error which have cumulative effects, in addition to the cumulative memory decay from decays in each of the saccade vector memories in the time since the saccade.

Table 1. Hypotheses expressed as seven competing models.

Index	Model name	Components
1	Allocentric	Allocentric Encoding Error + Allocentric Decay
2	Dual	Radial Encoding Error + Radial Decay + Angular Encoding Error + Angular Decay
3	Dual (Fixation)	Dual + Fixation Decay
4	Dual (Saccade)	Dual + Saccade Encoding Error + Saccade Decay
5	Dual + Interference	Dual + Interference
6	Dual (Fixation) + Interference	Dual + Fixation Decay + Interference
7	Dual (Saccade) + Interference	Dual + Saccade Encoding Error + Saccade Decay + Interference

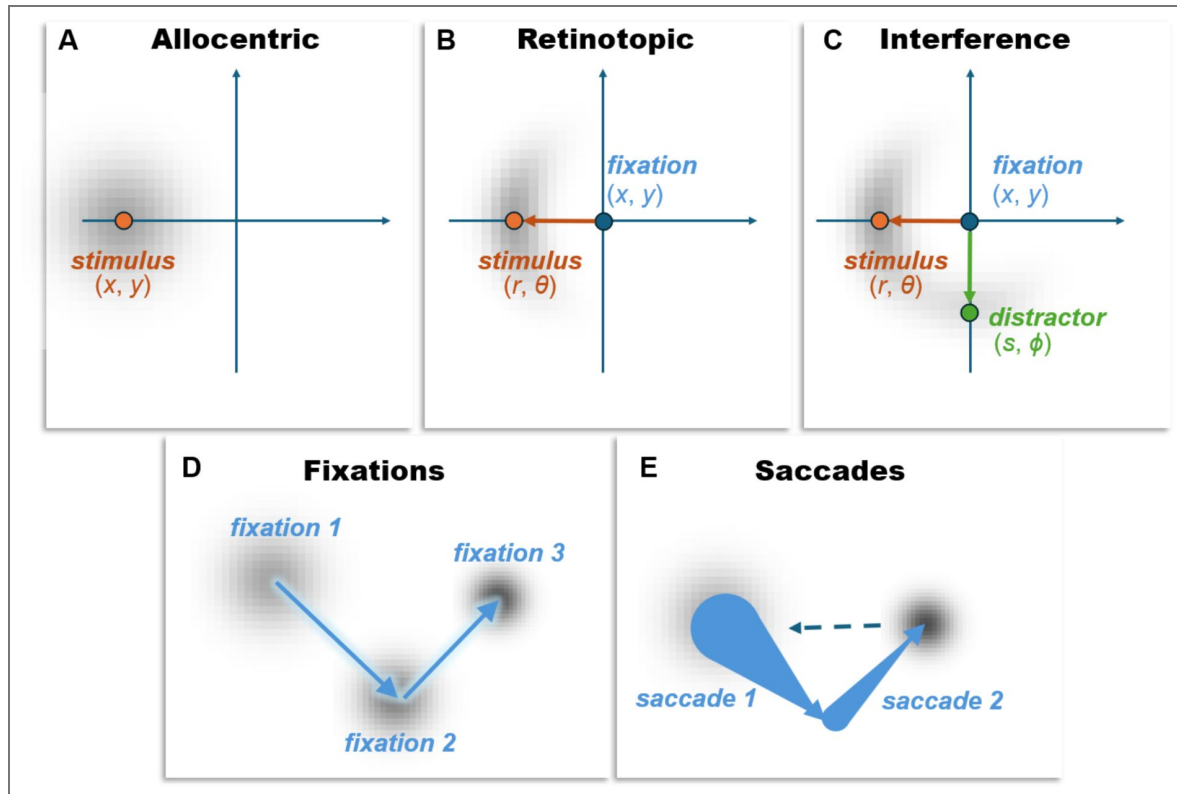


Figure 3. Conceptual illustration of allocentric vs retinotopic spatial representations.

This figure depicts the distinction between various representations of spatial memory. These include allocentric (world-centred) and retinotopic (eye-centred) spatial representation. **(A)** The **Allocentric** panel illustrates a world-centred reference frame, where a stimulus location (x, y) is represented independently of eye position. The spread of the grey shaded area around the stimulus indicates for this representation the encoding error and subsequent decay since the stimulus was viewed. In this framework, memory error is assumed to increase over time due to a general decay of the stored coordinates, which is treated as a decay in Euclidean space, such that the probability density of a remembered location (coordinates (x, y)) is a normal distribution whose covariance grows with time centred on the stimulus. This model corresponds to Model 1 in Table 1. **(B)** The **Retinotopic** panel presents an eye-centred reference frame, where the stimulus location is encoded relative to the current fixation point (x, y) using polar coordinates: radial distance (r) and angular direction (θ) . The orange arrow indicates the vector from fixation to the stimulus, representing these encoded retinotopic coordinates. The grey shaded area here again represents the initial encoding error and subsequent decay, which can be decomposed into radial and angular components. This **dual model**, incorporating both encoding error and decay in radial and angular components, corresponds to Model 2 in Table 1. It forms the basis for more complex models (Models 3-7 in Table 1) that differentiate between decay of the reference frame itself (e.g., recall of fixation point) and updating based on saccade information. **(C)** The **Interference** panel shows the combination of a stimulus and distractor (translated into the same retinotopic coordinates), with some probability associated with a response to the memory of the distractor stimulus. The lower two panels distinguish between whether the origin of the retinotopic plot is recalled based upon a remembered fixation location **(D)**, or whether it is based upon a reconstruction after propagating back through remembered saccade vectors **(E)** —conceptually shown with arrows whose base is large to indicate the (cumulative) uncertainty in the origin points of each vector.

5. **Dual + interference model:** This model tests the influence of cognitive load by incorporating crosstalk between items in memory. It suggests that the memory for the target location is partially corrupted by the presence of the other, distracting item's location, adding a source of error independent of remapping. In effect, this means the response distribution is assumed to be a weighted sum between the distributions anticipated if responding to the target or to the distractor.
6. **Dual (fixation decay) + interference model:** This model combines two potential sources of error: the passive decay of the remembered fixation point (Fixation Decay) and corruption from the distractor item (Interference).
7. **Dual (saccade update) + interference model:** This is the most complex model, combining the active, saccade-based updating mechanism with distractor interference. It proposes that memory error arises from both an imperfect remapping process based on a noisy saccade signal and from confusion with the other item held in memory.

By fitting these models to the trial-by-trial response data from all healthy participants ($N=42$), we adjudicated between competing mechanisms to determine which best explained participant performance (Figure 4 [↗](#)). We used random-effects Bayesian model selection to identify the most plausible generative model. This process relies on the marginal likelihood (model evidence), which inherently balances model fit against complexity—a principle often referred to as Occam's razor.^{26–28} The analysis yielded a strong result: the “Dual (Saccade) + Interference” model (Model 7 in Table 1 [↗](#)) emerged as the winning model, providing substantial evidence against the next best alternative with a Bayes Factor of 6.11. This indicates substantial evidence that saccade-based updating with distractor interference is the most prevalent generative process in the healthy population.

The superiority of this specific model provides three critical insights. Firstly, the victory of a dual-frame model over the simpler allocentric model strongly supports the hypothesis that spatial memory is fundamentally reliant on eye-centred representations that must be actively updated. Secondly, the model comparison decisively adjudicates the source of remapping error: the clear win for the transsaccadic updating mechanism suggests that transsaccadic error arises not from a passive forgetting of where the eyes were, but from an active, yet imperfect, process of remapping based on a noisy internal copy of the saccade command—or, perhaps, from the proprioceptive signal associated with that movement. Finally, the inclusion of the “Interference” term was essential for the winning model, demonstrating that this remapping process is also susceptible to confusion from other items held in memory. This winning model was therefore used for all subsequent analyses.

Patients with AD and PD show preserved saccadic interference but greater overall error

Having established the mechanisms in healthy individuals, we next investigated how these processes are affected in patients with Alzheimer's disease (AD) and Parkinson's disease (PD), comparing their performance to that of both young (YC) and elderly (EC) controls. As anticipated, there was a significant main effect of group on overall memory performance. A two-way ANOVA revealed that patient groups were less accurate on both location ($F(3, 318) = 59.71, p < 0.001$) and colour memory ($F(3, 318) = 47.85, p < 0.001$). Post-hoc analyses confirmed that AD patients exhibited significantly greater errors than all other groups across every condition (e.g., on No-Saccade Location Error vs YC, EC, and PD: all $p < 0.001$; see Figure 5 [↗](#)).

Across all groups, saccades selectively disrupted spatial memory—shown by a significant main effect of condition on location error ($F(3, 318) = 4.64, p = 0.003$)—but not on colour error ($F(3, 318) = 0.80, p = 0.49$). Bayesian repeated measures ANOVAs further supported this dissociation, providing moderate evidence for the null hypothesis in the AD group ($BF_{01} = 3.35, P(\text{excl} | \text{data}) = 0.77$) and weak evidence in the PD group ($BF_{01} = 2.23, P(\text{excl} | \text{data}) = 0.69$). This indicates that even in populations with established neurodegeneration, the detrimental impact of eye movements is specific to the spatial domain.

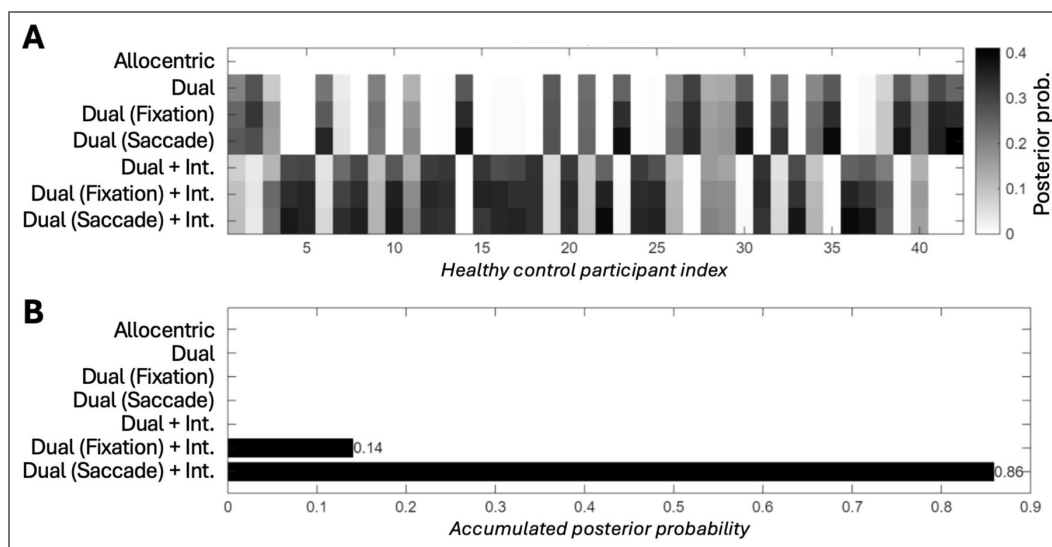


Figure 4. Model comparison results in healthy control (HC) participants (N=42).

(A) Individual participant posterior probabilities. Heatmap illustrates the posterior probability for each of the seven competing models across individual healthy participants. This panel indicates that the ‘Dual (Saccade Update) + Interference’ model frequently shows the highest posterior probability for individual participants. **(B)** Aggregate model posterior probabilities. Bar chart showing the posterior probability for each competing model, aggregated across all healthy participants.

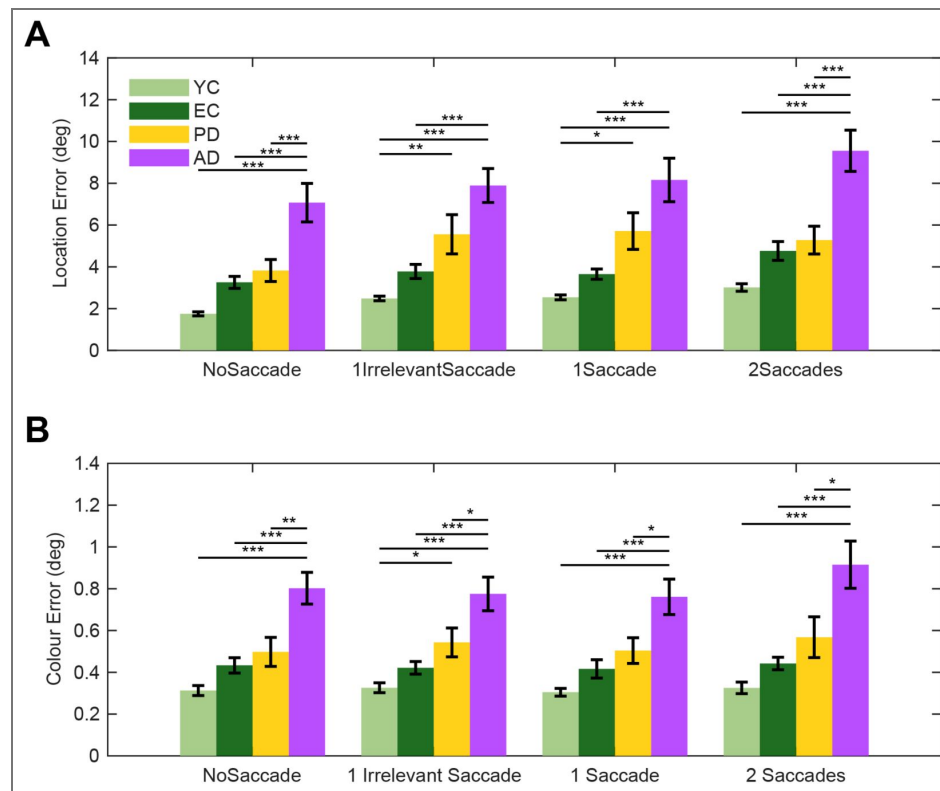


Figure 5. Saccadic interference with spatial memory is preserved across healthy aging and neurodegenerative conditions, despite group differences in recall error.

Saccades lead to increased recall error for location, but not colour, in healthy young (YC) and elderly (EC) controls, as well as in patients with PD and AD. While patient groups exhibit greater overall memory errors, the fundamental pattern of saccadic disruption remains consistent across all cohorts. **(A)** Location error. Mean location error (in degrees of visual angle) for each saccade condition (No-Saccade, 1 Irrelevant Saccade, 1 Saccade, 2 Saccades) for all participant groups. A two-way ANOVA revealed a significant main effect of Group and Condition, but there was no significant Group $\times$ Condition interaction ($F(9, 318) = 0.46$, $p = 0.900$), indicating that the magnitude of saccadic disruption to spatial memory is constant across groups. Post-hoc Tukey-Kramer comparisons within each condition between groups confirmed a graded impairment in overall spatial accuracy: AD patients exhibited significantly greater errors than all other groups (e.g., AD vs. YC, EC, PD: all $p < 0.001$), PD patients performed worse than both YC and EC (all $p < 0.001$), and EC performed worse than YC (all $p < 0.001$). The significant post-hoc between group comparisons are shown above the bar plots, with Bonferroni multiple comparison correction applied. **(B)** Colour error. Mean colour error (in radians) for each saccade condition across participant groups. For colour memory, a two-way ANOVA revealed a significant main effect of Group but no significant main effect of Condition or Group $\times$ Condition interaction. Post-hoc comparisons for colour error showed a similar pattern of overall group differences as observed for location memory (all group comparisons $p < 0.001$). Error bars represent ± 1 standard error of the mean (SEM). Asterisks denote statistical significance from post-hoc comparisons: *: $p < 0.05$; **: $p < 0.01$; ***: $p < 0.001$. A version of this figure displaying full data distributions via boxplots is provided in Supplementary Figure 2.

Most strikingly, the fundamental pattern of saccadic interference seen in healthy controls was fully replicated in the patient cohorts. In other words, the magnitude of this saccadic disruption did not differ between groups. We found no significant interaction between group and condition for either location ($F(9, 318) = 0.46, p = 0.90$) or colour memory ($F(9, 318) = 0.26, p = 0.98$). This critical null-interaction indicates that while overall spatial precision is lower in patients, the additional computational cost imposed by a saccade is not disproportionately larger.

The pattern of performance decay from the no-saccade to the two-saccade condition was remarkably consistent across all groups (Figure 5A), suggesting that the core mechanisms for updating spatial representations across eye movements are surprisingly resilient to the neurodegenerative processes in both PD and AD.

Modelling reveals the sources of spatial memory deficits in healthy aging and neurodegeneration

To understand the source of the observed deficits, we applied the winning ‘Dual (Saccade) + Interference’ model the data from all participants (YC, EC, AD, and PD). By fitting the model to the entire dataset, we obtained estimates of the parameters for each individual, which then formed the basis for our group-level analysis. To formally test for group differences, we used Parametric Empirical Bayes (PEB), a hierarchical Bayesian approach that compares parameter estimates across groups while accounting for the uncertainty of each estimate²⁹. This allowed us to identify which specific cognitive mechanisms, as formalised by the model parameters, were affected by age and disease.

The Bayesian inversion used here allows us to quantify the posterior mode and variance for each parameter and the covariance for each parameter. From these, we can compute the probabilities that pairs of parameters differ from one another, which we report as $P(A > B)$ —meaning the posterior probability that the parameter for group A was greater than that for group B.

We first examined the specific parameters differentiating healthy elderly (EC) from young controls (YC) to isolate the factors contributing to non-pathological, age-related decline. The analysis revealed that healthy ageing is primarily characterised by a significant increase in Radial Decay ($P(EC > YC) = 0.995$), a heightened susceptibility to Interference ($P(EC > YC) = 1.000$), and a reduction in initial Angular Encoding precision ($P(YC < EC) = 0.002$; Figure 6). These results suggest that normal ageing degrades the fidelity of the initial memory trace and its resilience over time, while the core computational process of updating information across saccades remains intact.

Beyond these baseline ageing effects, our clinical cohorts exhibited more severe and condition-dependent impairments. Radial decay showed a clear, graded impairment: AD patients had a greater decay rate than PD patients ($P(AD > PD) = 1.000$), who in turn were more impaired than the EC group ($P(PD > EC) = 0.996$). A similar graded pattern was observed for Interference, where AD patients were most susceptible ($P(AD > PD) = 0.999$), while the PD and EC groups did not significantly differ ($P(PD > EC) = 0.532$).

Patients with AD also showed a tendency towards greater angular decay than controls ($P(AD > EC) = 0.772$), although this fell below the 95% probability threshold. This effect was influenced by a lower decay rate in the PD group compared to the EC group ($P(PD < EC) = 0.037$). In contrast, group differences in encoding were less pronounced. While YC exhibited significantly higher precision than all other groups, AD patients showed significantly higher angular encoding error than PD patients ($P(AD > PD) = 0.985$), though neither group differed significantly from the EC group.

Crucially, parameters related to the saccade itself—saccade encoding and saccade decay—did not differentiate the groups. This indicates that neither healthy ageing nor the early stages of AD and PD significantly impair the fundamental machinery for transsaccadic remapping. Instead, the visuospatial deficits in these conditions arise from specific mechanistic failures: a faster decay of radial position information and increased susceptibility to interference, both of which are present in healthy ageing but significantly amplified by neurodegeneration.

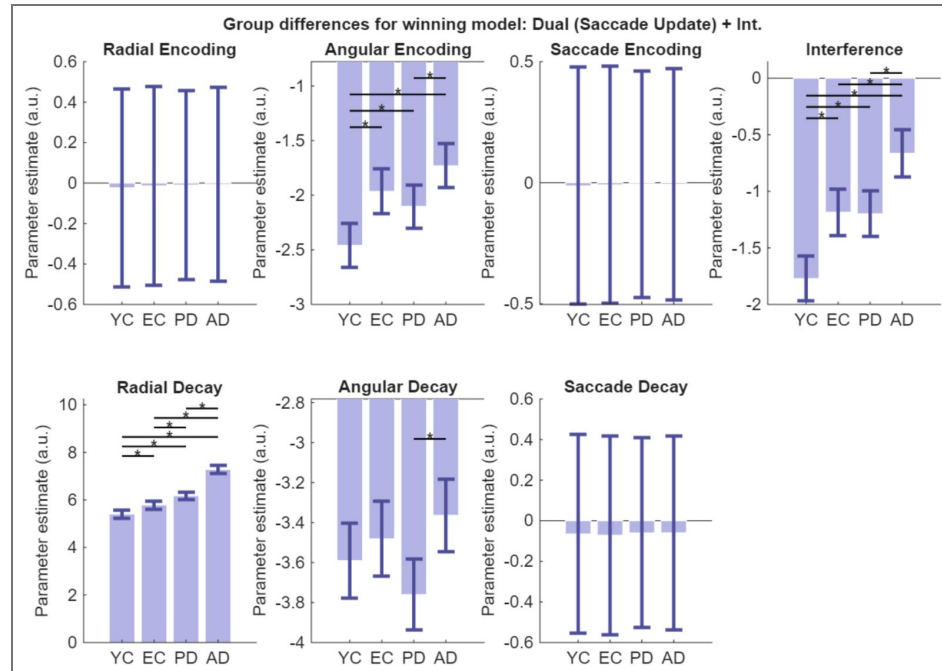


Figure 6. Group differences in winning model parameters

The plots show group-level parameter estimates from the Parametric Empirical Bayes (PEB) analysis for the winning ‘Dual (Saccade) + Interference’ model. The y-axis represents the posterior mean of the log-scaled parameter, where a more positive value indicates a greater influence of that parameter on performance (i.e., greater error or faster decay). The exception is the interference parameter, which is effectively a logit parameter, meaning the probability of selecting the distractor is obtained by a sigmoid transform. Error bars represent the 90% posterior confidence intervals. Group differences are indicated by an asterisk (*) if the posterior probability of a difference between the respective parameter estimates is greater than 95%, as calculated from their posterior distributions (i.e., if $P(A > B) > 0.95$ or $P(B > A) > 0.95$). The analysis reveals that Angular Encoding, Angular Decay, Radial Decay, and Interference parameters differentiate the groups. Critically, the EC group shows significant deviations from the YC group in Angular Encoding, Radial Decay, and Interference, providing a model-based profile of healthy age-related spatial memory decline. Notably, parameters related to saccade processing (Saccade Encoding and Saccade Decay) and Radial Encoding do not differ between groups, suggesting that the integrity of transsaccadic remapping is preserved across the lifespan and in early-stage neurodegeneration.

Transsaccadic working memory predicts clinical constructional deficits

To assess whether the mechanistic parameters derived from the LOCUS task represent core phenotypes of real-world visuospatial abilities, we also instructed all participants to complete the Rey-Osterrieth Complex Figure copy task (ROCF; [Figure 7A](#)) on an Android tablet using a digital pen (see examples in [Figure 7B](#); all Copy data are available in the open dataset: <https://osf.io/95ecp/>). The ROCF is a gold-standard neuropsychological tool for identifying constructional apraxia.³⁰ Historically, drawing performance has shown only weak or indirect correlations with traditional working memory measures.³¹ This disconnect has been attributed to active visual-sampling strategies—frequent eye movements that treat the environment as an external memory buffer, minimising the necessity of holding large volumes of information in internal working memory.^{3–5}

We hypothesised that drawing accuracy is primarily constrained by the precision of spatial updating across frequent saccades rather than raw memory capacity. To evaluate the ecological validity of the identified saccade-updating mechanism, we modelled individual ROCF copy scores across all four groups using the estimated (maximum a posteriori) parameters from the winning “Dual (Saccade) + Interference” model (Model 7; [Figure 8](#)) as regressors in a Bayesian linear model. Prior to inclusion, each regressor was normalised by dividing by the square root of its variance.

This model successfully explained 61.99% of the variance in ROCF copy scores, indicating that these computational parameters are strong predictors of real-world constructional ability ([Figure 8A](#)). The posterior covariance matrix, illustrating the regression coefficients associated with each of the model parameters, is shown in [Figure 8B](#). [Figure 8C](#) indicates how specific mechanistic parameters during transsaccadic working memory contributed to explaining ROCF copy performance. Although as shown in [Figure 6](#) the saccade encoding error did not differentiate across these four groups, it does contribute to a worse ROCF copying performance. This highlights the critical role of accurate remapping based on saccadic information; even if the core saccadic update mechanism is preserved across groups (as shown in previous analyses), the precision of this updating process is crucial for complex visuospatial tasks. Moreover, worse ROCF copy performance is associated particularly with higher initial angular encoding error. This indicates that imprecision in the initial registration of angular spatial information contributes to difficulties in accurately reproducing complex visual stimuli.

Surprisingly, less angular decay was associated with poorer ROCF copy scores. However, a plausible explanation for this could be related to the trend seen in angular encoding error. People who have smaller angular encoding error have more room for memory decay before angular encoding error accumulates. Evidence for this explanation is evident in the (weak) negative correlation evident in the covariance matrix in [Figure 8](#) between the coefficients for angular encoding and decay rates. The implication is that the results would be similar if we slightly increased the encoding variance while reducing the decay rate, and vice versa.

Conversely, parameters such as radial encoding error, radial decay, saccade decay and interference did not predict ROCF copy scores. This differential predictive power underscores the specific relevance of angular and saccade-related precision for the demanding task of copying complex figures, which inherently requires continuous and accurate spatial updating across multiple fixations.

Furthermore, to evaluate the specificity of the relationship between transsaccadic updating and constructional ability, we conducted a partial correlation analysis. The association between the transsaccadic working memory in LOCUS and ROCF copy performance remained highly significant ($\rho = -0.46$, $p < 0.001$), even after accounting for age, education, and general memory performance (ACE-III Memory subscore).

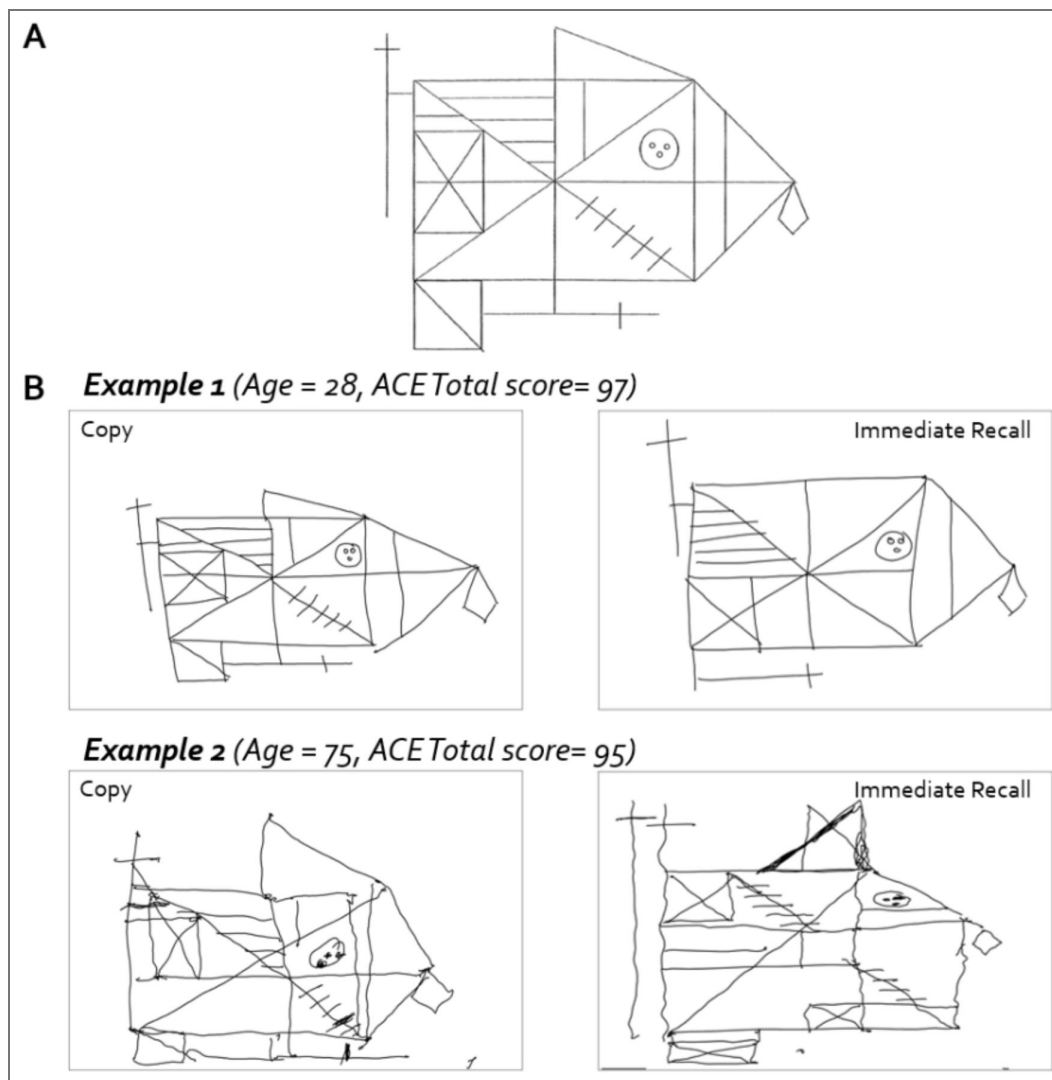


Figure 7. Rey-Osterrieth Complex Figure (ROCF) copying performance.

(A) The original figure that participants copied and recalled. (B) Examples of ROCF copy and immediate recall tasks from a young and an elderly healthy participant, both with normal cognitive function (ACE total score > 88). ACE = Addenbrooke's Cognitive Examination-III.

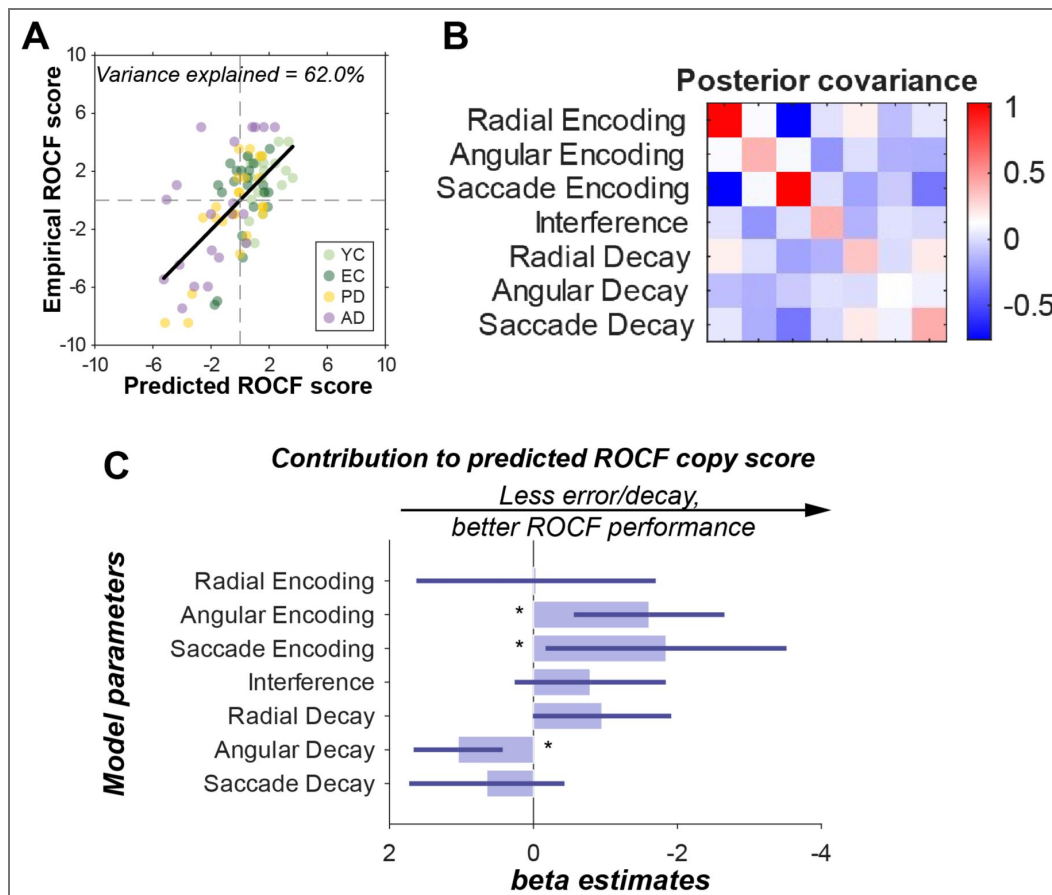


Figure 8. Effect of model parameters on Rey-Osterrieth Complex Figure copy scores.

How model parameters from the winning “Dual (Saccade) + Interference” model explain variance in ROCF copy performance? (A) Predicted versus empirical scores. A scatter plot of empirical ROCF scores against scores predicted by the fitted linear model. Each point represents an individual participant, coloured by diagnostic group: young controls (YC; light green), elderly controls (EC; dark green), Parkinson’s disease (PD; yellow), and Alzheimer’s disease (AD; purple). A black line denotes the linear fit across all data. ROCF scores were demeaned across participants. Scores outside of 2 standard deviations from the mean were omitted. The plot title indicates the percentage of variance in ROCF copy scores explained by the model. (B) Posterior covariance matrix. A heatmap depicting the posterior covariance between the estimated model parameters (i.e., between the regression coefficients that determine the contribution of each of the parameters from our previous winning model to the ROCF score). Diagonal elements represent individual parameter variances; off-diagonal elements show covariances between parameter pairs. The colour bar provides the scale. (C) Estimated parameter contributions. Bar plot displaying the estimated contribution (beta coefficients) of each LOCUS model parameter to the predicted ROCF copy score. Error bars represent 90% credible intervals (1.65 standard deviations of the posterior variance). Asterisks (*) denote parameters whose 90% confidence interval does not encompass zero.

Discussion

This study introduced a novel paradigm, the LOCUS task ([Figure 1](#)), and combined this with computational modelling to investigate the underlying mechanisms and sources of potential errors in dynamic spatial working memory, particularly how the brain maintains a stable world-centred frame of reference across saccades. We investigated this across four cohorts: young healthy adults (YC), healthy elderly adults (EC), and patients with and Alzheimer’s disease (AD) and Parkinson’s disease (PD). The two clinical groups are known to be frequently associated with significant disruptions to visuospatial processing and oculomotor control.^{21,22}

The LOCUS task was designed to dissociate retinotopic (within-fixation) and transsaccadic (across-fixation) working memory, examining the precision of both location and colour information maintained across saccades. Our findings reveal a “saccade cost”—a reduction in spatial memory precision following eye movements, consistent with previous research on healthy young adults.²³ Unexpectedly, this saccade cost was remarkably consistent across all four populations, demonstrating its stability with aging and neurodegeneration. This finding implies that any observed reduction in transsaccadic performance in older adults and neurodegenerative patients is primarily attributable to lower baseline retinotopic memory rather than an increased cost associated with saccadic remapping.

The conclusion was further supported by our computational modelling. The winning model, “Dual (Saccade) + Interference”, posits an eye-centred, retinotopic representation (encoding stimulus as angular and radial displacement from fixation) that also incorporates saccade-based updating and interference. Although saccade updating was an essential component of the winning model, its two key parameters—initial encoding error and decay rate during maintenance—did not significantly differ across groups. This indicates that the core computational process of updating spatial information based on eye movements is largely preserved in healthy aging and neurodegeneration.

Instead, group differences were driven by deficits in angular encoding error (precision of initial angle from fixation), angular decay, radial decay (decay in memory of distance from fixation), and interference susceptibility. This implies a functional and neuroanatomical dissociation: while the ventral stream (the “what” pathway) shows an age-related decline in the quality and stability of stored representations, the dorsal-stream (the “where” pathway) parietal-frontal circuits responsible for coordinate transformations remain functionally robust.^{32–35} These spatial updating mechanisms appear resilient to the normal ageing trajectory and only break down when challenged by the specific pathological processes seen in Alzheimer’s or Parkinson’s disease.

The eye-centred retinotopic representation identified by our model is consistent with findings in posterior parietal cortex neurons^{34,35} which are considered to play a crucial role in visual localisation as well as remapping across eye movements.^{9,36} In using this computational approach, we align our work with established practices in computational psychiatry: Our framework employs Variational Laplace, a method used to recover computational phenotypes in clinical populations like those with substance use disorders,^{37,38} and the models we fit using this procedure feature time-dependent parameterisation of variance—conceptually similar to the widely-used Hierarchical Gaussian Filter.^{39–42} While we describe the present framework as ‘computational’, it is more precisely characterised as an algorithmic-level generative model within Marr’s hierarchy. Our focus was on defining the rules of spatial integration and the sources of eye-movement-induced noise, rather than deriving these processes from normative principles or defining their specific neural implementation. Importantly, the risk of overfitting is mitigated by the Bayesian Model Selection framework; by utilising the marginal likelihood for model comparison, the procedure inherently penalises excessive model complexity and promotes generalisability.^{26–28,43} This generalisability was further evidenced by the model’s ability to predict performance on the independent ROCF task, confirming that these parameters represent robust mechanistic phenotypes rather than idiosyncratic fits to the initial dataset.

A clear finding was the specificity of the saccade cost to spatial features; it was not observed for non-spatial features like colour, even in neurodegenerative conditions. This discrepancy challenges notions of fixed visual working memory capacity unaffected by saccades.^{16,44–46} The differential impact on spatial versus non-spatial features in transsaccadic memory aligns with the established “what” and “where” pathways in visual processing.^{32,33} For objects to remain unified, object features must be bound to stable representations of location across saccades.¹⁹ One possibility is that remapping updates both features and location through a shared mechanism, predicting equal saccadic interference for both colour and location in the present study.

However, our findings suggest otherwise. One potential concern is whether this dissociation simply reflects the inherent spatial noise introduced by fixational eye movements (FEMs), such as microsaccades and drifts.⁴⁷ Because locations are stored in a retinotopic frame, fixational instability necessarily shifts retinal coordinates over time. However, the “saccade cost” here was defined as the error increase relative to a no-saccade baseline of equal duration; because both conditions are subject to the same fixational drift, any FEM-induced noise is effectively subtracted out. Thus, despite the ballistic and non-Gaussian nature of FEMs,⁴⁸ they cannot account for the fact the saccade cost in the spatial memory, but total absence in the colour domain. Another possibility is that this dissociation reflects differences in baseline task difficulty or dynamic range. Yet, the presence of a robust recency effect in colour memory (Figure 2B [↗](#)) confirms that our paradigm was sensitive to memory-dependent variance and was not limited by floor or ceiling effects.

The fact that identical eye movements—executed simultaneously and with identical vectors—systematically degraded spatial precision while sparing colour suggests a feature-specific susceptibility to transsaccadic remapping. This supports the view that the computational process of updating an object’s location involves a vector-subtraction mechanism—incorporating noisy oculomotor commands (efference copies)—that introduces specific spatial variance. Because this remapping is a coordinate transformation, the resulting sensorimotor noise does not functionally propagate to non-spatial feature representations. Consequently, features like colour may be preserved or automatically remapped without the precision loss associated with spatial updating.^{11,49} Our paradigm thus provides a refined tool to investigate the architecture of transsaccadic working memory across distinct object features.

Importantly, our computational framework establishes a direct mechanistic link between transsaccadic updating and real-world constructional ability. Specifically, higher saccade and angular encoding errors contribute to poorer ROCF copy scores. By mapping these mechanistic estimates onto clinical scores, we found that the parameters derived from our winning model explain approximately 62% of the variance in constructional performance across groups. These findings suggest that the computational parameters identified in the LOCUS task represent core phenotypes of visuospatial ability, providing a mechanistic bridge between basic cognitive theory and clinical presentation.

This relationship provides novel insights into the cognitive processes underlying drawing, specifically highlighting the role of transsaccadic working memory. Previous research has primarily focused on the roles of fine motor control and eye-hand coordination in this skill.^{4,50–55} This is partly because of consistent failure to find a strong relation between traditional memory measures and copying ability.^{4,31} For instance, common measures of working memory, such as digit span and Corsi block tasks, do not directly predict ROCF copying performance.^{31,56} Furthermore, in patients with constructional apraxia, these memory performance measures often remain relatively preserved despite significant drawing impairments.^{56–58} In the literature, this lack of association has often been attributed to “deictic” visual-sampling strategies, characterised by frequent eye movements that treat the environment as an external memory buffer, thereby minimising the need to maintain a detailed internal representation.^{4,59} In a real-world copying task, the ROCF requires a high volume of saccades, making it uniquely sensitive to the precision of the dynamic remapping signals identified here. Recent eye-tracking evidence confirms that patients with AD exhibit significantly more saccades and longer fixations during figure copying compared to controls, potentially as a compensatory response to transsaccadic working memory

constraints.⁵⁶ This high-frequency sampling—averaging between 150 and 260 saccades for AD patients compared to approximately 100 for healthy controls—renders the task highly dependent on the precision of dynamic remapping signals.⁵⁶ To ensure this relationship was not driven by a general “g-factor” or non-spatial memory impairment, we further investigated the role of broader cognitive performance using the ACE-III Memory subscale. We found that the relationship between transsaccadic working memory and ROCF performance remains highly significant, even after controlling for age, education, and ACE-III Memory subscore. This suggests that transsaccadic updating may represent a discrete computational phenotype required for visuomotor control, rather than a non-specific proxy for global cognitive decline.⁵⁸

In other words, even when visual information is readily available in the world, the act of copying depends critically on working memory across saccades. This reveals a fundamental computational trade-off: while active sampling strategies (characterised with frequent eye-hand movements) effectively reduce the load on capacity-limited working memory, they simultaneously increase the demand for precise spatial updating across eye movements. By treating the external world as an “outside” memory buffer, the brain minimises the volume of information it must hold internally, but it becomes entirely dependent on the reliability with which that information is remapped after each eye movement. This perspective aligns with, rather contradicts, the traditional view of active sampling, which posits that individuals adapt their gaze and memory strategies based on specific task demands.^{3,60} Furthermore, this perspective provides a mechanistic framework for understanding constructional apraxia; in these clinical populations, the impairment may not lie in a reduced memory “span,” but rather in the cumulative noise introduced by the constant spatial remapping required during the copying process.^{58,61}

Beyond constructional ability, these findings suggest that the primary evolutionary utility of high-resolution spatial remapping lies in the service of action rather than perception. While spatial remapping is often invoked to explain perceptual stability,^{11–13,15} the necessity of high-resolution transsaccadic memory for basic visual perception is debated.^{13,62–64} A prevailing view suggests that detailed internal models are unnecessary for perception, given the continuous availability of visual information in the external world.^{13,44} Our findings support an alternative perspective, aligning with the proposal that high-resolution transsaccadic memory primarily serves action rather than perception.¹³ This is consistent with the need for precise localisation in eye-hand coordination tasks such as pointing or grasping.⁶⁵ Even when unaware of intrasaccadic target displacements, individuals rapidly adjust their reaching movements, suggesting direct access of the motor system to remapping signals.⁶⁶ Further support comes from evidence that pointing to remembered locations is biased by changes in eye position,⁶⁷ and that remapping neurons reside within the dorsal “action” visual pathway, rather than the ventral “perception” visual pathway.^{13,68,69} By demonstrating a strong link between transsaccadic working memory and drawing (a complex fine motor skill), our findings suggest that precise visual working memory across eye movements plays an important role in complex fine motor control.

Materials and Methods

Ethics

The study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments. Ethical approval was granted by the University of Oxford ethics committee (IRAS ID: 248379, Ethics Approval Reference: 18/SC/0448). All participants gave written informed consent prior to the start of the study.

Participants

A total of 87 participants completed the study: 21 young healthy adults (YC), 21 older healthy adults (EC), 23 patients with Parkinson’s disease (PD), and 22 patients with Alzheimer’s disease (AD). Their demographic and clinical details are summarised in Table 2 [↗](#). Initially, 90 participants were recruited (22 YC, 21 EC, 25 PD, 22 AD); however, three individuals (1 YC and 2 PD) were excluded from all analyses due to technical issues during data acquisition.

Table 2. Demographics of Participants.

The value in brackets indicate one standard deviation or proportion of the sample. YC = Young healthy control; EC = Elderly healthy control; PD = Parkinson's Disease; AD = Alzheimer's Disease; ACE = the Addenbrooke's Cognitive Examination III. Two relevant ACE-III subscores are also presented (Memory and Visuospatial).

Group	N	Age	Gender (N of Females)	Handedness (N of Right handed)	Education (Years)	ACE Total	ACE Memory	ACE Visuospatial
YC	21	24.1 (4.7)	15 (71.4%)	18 (85.7%)	17.0 (2.0)	96.6 (2.9)	25.0 (1.2)	15.7 (0.6)
EC	21	72.4 (6.3)	12 (57.1%)	18 (85.7%)	16.6 (4.9)	96.1 (3.5)	24.0 (2.9)	15.8 (0.7)
PD	23	70.8 (6.2)	11 (47.8%)	21 (91.3%)	15.7 (3.4)	93.7 (4.9)	24.2 (2.6)	15.3 (0.9)
AD	22	68.2 (9.7)	9 (40.9%)	16 (72.7%)	14.6 (4.1)	78.3 (13.0)	17.8 (5.3)	13.5 (2.2)

All participants were recruited locally in Oxford, UK. None were professional artists, had a history of psychiatric illness, or were taking psychoactive medications (excluding standard dopamine replacement therapy for PD patients). Young participants were recruited via the University of Oxford Department of Experimental Psychology recruitment system. Older healthy volunteers (all >50 years of age) were recruited from the Oxford Dementia and Ageing Research (OxDARE) database.

Patients with PD were recruited from specialist clinics in Oxfordshire. All had a clinical diagnosis of idiopathic Parkinson's disease and no history of other major neurological or psychiatric conditions. While specific dosages of dopamine replacement therapy (e.g., levodopa equivalent doses) were not systematically recorded, all patients were tested while on their regular medication regimen ('ON' state).

Patients with PD were recruited from clinics in the Oxfordshire area. All had a clinical diagnosis of idiopathic Parkinson's disease and no history of other major neurological or psychiatric illnesses. While all patients were tested in their regular medication 'ON' state, the specific pharmacological profiles—including the exact types of medication (e.g., levodopa, dopamine agonists, or combinations) and dosages—were not systematically recorded. The disease duration and PD severity were also un-recorded for this study.

Patients with AD were recruited from the Cognitive Disorders Clinic at the John Radcliffe Hospital, Oxford, UK. All AD participants presented with a progressive, multidomain, predominantly amnesic cognitive impairment. Clinical diagnoses were supported by structural MRI and FDG-PET imaging consistent with a clinical diagnosis of AD dementia (e.g., temporo-parietal atrophy and hypometabolism).⁷⁰ All neuroimaging was reviewed independently by two senior neurologists (S.T. and M.H.).

Global cognitive function was assessed using the Addenbrooke's Cognitive Examination-III (ACE-III).⁷¹ All healthy participants scored above the standard cut-off of 88, with the exception of one elderly participant who scored 85. In the PD group, two participants scored below the cut-off (85 and 79). In the AD group, six participants scored above 88; these individuals were included based on robust clinical and radiological evidence of AD pathology rather than their ACE-III score alone.

Experimental procedure

The LOCUS task (L^Ocation and CO^Lour Updating across Saccades, [Figure 1](#)) assessed visuospatial working memory across eye movements. Participants viewed two sequentially presented coloured squares (1-second duration each, with a 0.5-second inter-stimulus interval), memorising their location and colour within frames centered on a fixation cross. A labelled colour wheel then cued recall of a specific square. Participants selected the remembered colour and dragged a corresponding square to its original screen location.

Saccade demands were manipulated by varying the location of the second frame and colour wheel. In 50% of trials, the second frame appeared in the same location as the first (no saccade required), while in the other 50%, it appeared in a different location, requiring a saccade of at least 8.5° (degrees of visual angle). The colour wheel also appeared in the same or a different location as the second frame with equal probability. This creates four conditions: No-saccade, Saccade-After-Item 1, Saccade-After-Item-2, or Two-Saccades ([Figure 1](#)). Each condition comprised 25% of the 160 trials, presented in 16 randomised blocks.

The coloured square was 0.66° × 0.66° visual angle, the fixation cross was 1° × 1° visual angle, and the frame was 9.8° × 9.8° visual angle with a 0.1° linewidth. The location of the square was randomised within the frame with a fixed minimal distance from fixation cross of 3.82° visual angle. The screen had 3 × 5 possible frame locations with a 0.3° gap between frames. The colour wheel was 4.5° wide, with a 1.5°-wide inner black circle displaying a 1° tall label (probing "1" or "2"). The colour and location of squares were generated pseudorandomly for each participant just before the starting experiment. To avoid the potential bias to certain locations on the colour wheel, the colour wheel's rotation was randomised for every trial.

Participants were instructed to maintain fixation on a cross when it was present, and to execute saccades to peripheral targets when prompted by appearance of a new frame at a different location or the appearance of the colour wheel. When item 1 or 2 was displayed, it was positioned within a square frame with a central fixation cross (Figure 2). The use of a square frame and central fixation cross served to emphasise the retinotopic reference frame, highlighting no need to move the eyes unless these elements were extinguished, and a new stimulus appeared in the periphery. Thus, in the no-saccade condition, participants did not need to make eye movements and therefore respond without having to rely on transsaccadic memory.

To mitigate potential confounds related to visual impairment or inability to fixate with task instructions, we monitored eye position throughout the experiment. Eye-tracking analysis confirmed high compliance: participants fixated the cross in the majority of trials ($81\% \pm 10\%$), with no significant difference between age groups ($t(40) = 0.76, p = 0.45$). Fixation durations averaged $0.21s (\pm 0.05s)$, sufficient for encoding visual information. Furthermore, in the no-saccade condition, participants successfully maintained fixation in 80% of trials ($\pm 10\%$). Similarly, in the saccade conditions, participants accurately executed saccades away when required on 86% of trials ($\pm 18\%$). Non-compliance trials were excluded from further analysis. Recall was performed in a fixed order, with colour reported before location. This sequence was primarily chosen to minimise cognitive load and task-switching demands for the two neurological patient cohorts, ensuring the paradigm remained accessible for individuals with AD and PD. While this order results in a slightly longer retention interval for location recall, the saccade cost was identified by comparing location error across experimental conditions with similar timings but varying saccadic demands.

Measurement of saccade cost

The LOCUS task allows for the quantification of several key metrics: (1) retinotopic working memory: the location error in no-saccade condition, (2) transsaccadic working memory: the location error in two-saccades condition, and (3) “saccade cost”: the reduction in spatial memory accuracy attributable to saccades.

Within this framework, it is important to distinguish between the broad construct of spatial working memory and the specific process of transsaccadic memory. While spatial working memory refers to the general ability to maintain spatial information over short intervals, transsaccadic memory describes the dynamic updating of these representations—termed remapping—to ensure stability across eye movements. Unlike classical ‘static’ measures of spatial working memory, such as the Corsi block task which focuses on memory span, transsaccadic memory tasks explicitly require the integration of stored visual information with motor signals from intervening saccades. Our paradigm treats transsaccadic updating as a core computational process within spatial working memory, where eye-centred representations are actively reconstructed based on noisy memories of the intervening saccade vectors.

While a straightforward approach to quantify saccade cost would be to directly compare performance in the two-saccades condition to the no-saccade condition, this method confounds saccade cost with retinotopic working memory. To isolate the effect of saccades on memory independent of retinotopic condition, we calculated saccade cost as the difference in performance between the Two-Saccade condition and the Saccade-After-The-Other-Item condition. This yields a specific measure of the additional memory error due to a saccade made immediately after the probed item.

Furthermore, we conducted a secondary analysis using the beta coefficient derived from a multiple linear regression model to estimate the effect of saccades and item order on location errors. This approach confirmed the relationship between saccade cost and ROCF copy performance, demonstrating the robustness of our findings.

Rey-Osterrieth Complex Figure (ROCF)

Participants completed the ROCF copy and immediate recall tasks on a Galaxy Tab S3 tablet (SM-T820) using the corresponding digital pen (Galaxy Tab S3 S Pen). The tablet's screen dimensions were 148 x 197 mm (1536 x 2048 pixels, 264 ppi density). Participants placed the tablet on the table upright. During the copy phase, the ROCF figure displayed on the top of the tablet⁷². They first copied the figure in the lower portion of the screen. Upon completion, they were instructed to reproduce the ROCF from memory, without the figure present. Drawing traces were recorded and printed for blind scoring by two experienced raters using the standard 18-element scoring system.⁷³

For further analysis, the drawing paths were analysed in MATLAB using in-house scripts. We extracted the total time taken to complete each task, calculated from the first moment the digital pen touched the screen to the final pen lift. In addition, we extracted total distance drawn (total path length in pixels), path drawing speed (mean derivative of path distance per second), and mean waiting time (duration between completing the last path and starting the next path).

Computational modelling

We used a series of simple computational models to test a set of hypotheses about the mechanisms that underwrite the measured data. All hypotheses rely upon the idea that memory decays with time since the stimulus was presented, modelled here using a diffusion (or Brownian motion) like process. Under this formulation, the shape of the distribution of expected responses, and its change over time, will depend upon the way in which our brains represent the variables that we expect to decay. Conceptually, our hypotheses distinguished between allocentric (screen centred) coordinates and retinotopic coordinates, whether target and distractor representations do or do not interfere with one another, whether there was evidence for an embodied representation in which the current position of the eyes helps reinforce a memory and, if so, whether remembered locations are centred on remembered fixations or on relative position based upon memories of the vectors of saccades performed since that location.

Figure 9⁷⁴ shows the form of these hypotheses, each of which can be expressed with a time-dependent likelihood function. Each of these models was fit to the responses made by each participant, where the response data were rotated and translated such that the target stimulus was always in the same angle, and the origin was always set to be the fixation location associated with the probed stimulus. The model fits were performed using a Variational Laplace procedure (a form of approximate Bayesian inference) using a Newton optimisation scheme as implemented in SPM25 using the `spm_nlsi_Newton.m` MATLAB routine.^{74,75} This relies upon a specification of a log likelihood function which gives the distribution, given the parameters, of the recorded responses. In the allocentric model, this log likelihood function had the form:

$$\mathcal{L}(\kappa) = \ln(\mathcal{N}(u; x, \kappa_1 \tau + \kappa_2) \times \mathcal{N}(v; y, \kappa_1 \tau + \kappa_2))$$

Here, (u, v) are the Euclidean coordinates of the response to a stimulus presented at coordinates (x, y) , with an encoding variance of κ_2 and a decay constant of κ_1 which scales with the time τ between the stimulus presentation and the response. This is summed over all responses a participant has made accounting for the time at which they are made to give the accumulated log likelihood for the parameters for that dataset.

The dual models, with a retinotopic component, all have the form:

$$\mathcal{L}(\kappa) = \ln\left(\mathcal{N}\left(\sqrt{(u-x)^2 + (v-y)^2}; r, \Sigma_r(\kappa, \tau, t, T)\right) \times \mathcal{N}\left(\tan^{-1}\left(\frac{v-y}{u-x}\right); \theta, \Sigma_\theta(\kappa, \tau, t, T)\right)\right)$$

Here, the principle is similar, but the distributions are over the polar coordinates of the response, with (r, θ) representing the polar coordinates of the stimulus. The variances, Σ , are functions now of the parameter set and (up to) three different time variables. These different times are illustrated schematically in Figure 9⁷⁴ (middle panel) and are disambiguated by the different experimental conditions. They are:

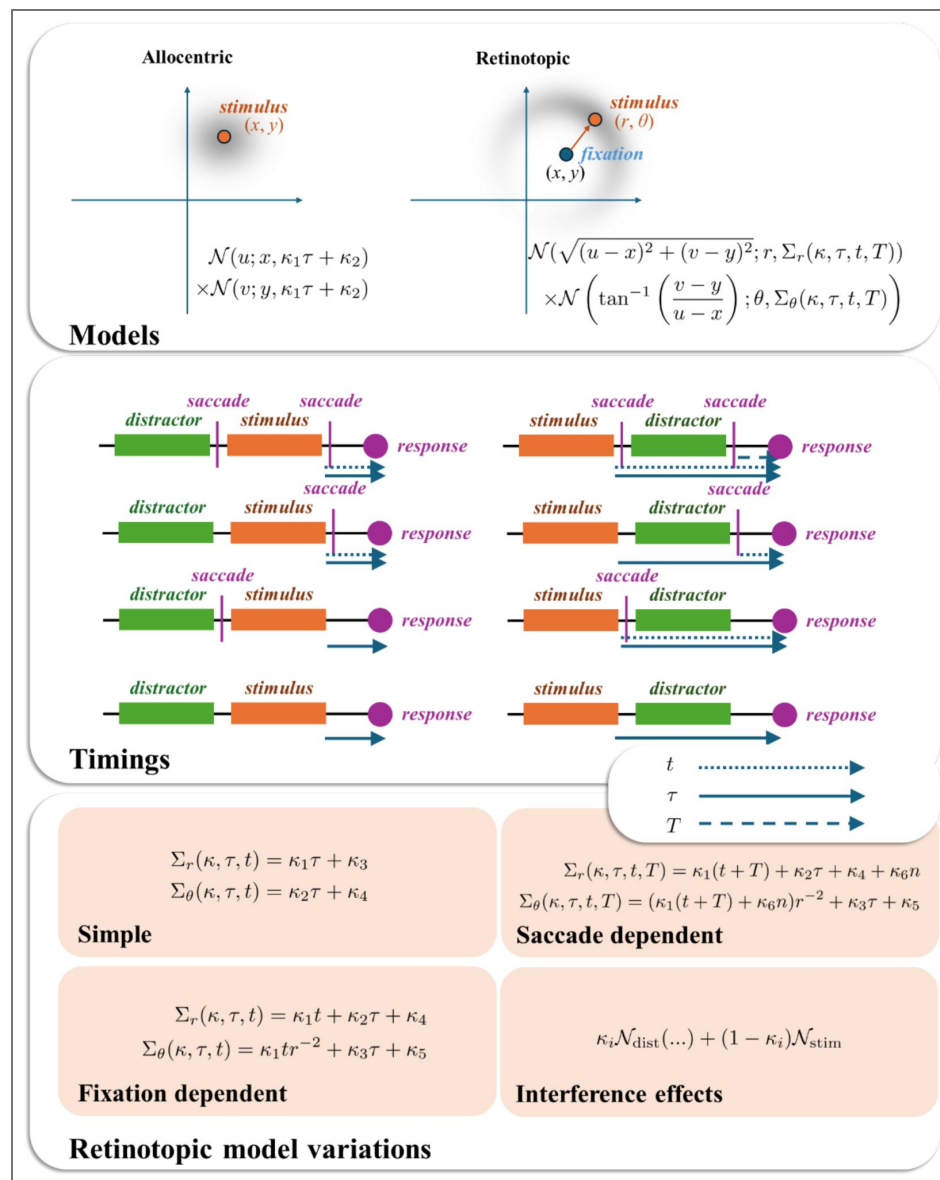


Figure 9. Models and hypotheses.

This figure illustrates the form of the models used for our model comparisons. The upper panel shows the distinction between allocentric screen centred coordinates, which we treat as a decay in Euclidean space, such that the probability density of a response (coordinates (u,v)) is a normal distribution whose covariance grows with time centred on the stimulus. In contrast, the retinotopic representation uses polar coordinates with both radial and angular decays. There are several ways in which the associated covariance can be formulated, and this depends upon the different times over which the decay in memory might occur. The different timings in the alternative experimental conditions are shown schematically in the middle panel, dealing with the time since the stimulus was visible (τ), the time since the first saccade (if any) since the stimulus was present (t), and the time since the second saccade (if any) after the stimulus (T). The variations, based on these timings, for the dynamics of the covariance for the retinotopic model, are shown in the lower panel. The simple model assumes that any decay in the fixation location around which the retinotopic coordinates can be captured by decays in the radial and angular coordinates. This implies there is no difference in the form of the decay when the same fixation location is sustained after the stimulus disappears compared to when a subsequent saccade is made. The fixation-dependent model assumes instead that there is an independent decay in the fixation location, which only begins once we have moved our eyes from the location at which we saw the stimulus. In contrast, the saccade dependent framing assumes that, rather than retain a memory of fixation location, we reconstruct the fixation location based upon our memory of the saccade vectors made, since that fixation. Here, the n variable represents the number of saccades made since the relevant fixation. The interference effects are mediated by computing the distribution we would expect from a response to the target stimulus and the distribution we would expect from a response to the distractor stimulus (translated into the same retinotopic space) and taking a weighted sum of the two.

1. the time since the stimulus was visible (τ)
2. the time since the first saccade (if any) since the stimulus was present (t)
3. the time since the second saccade (if any) after the stimulus (T).

The forms of the variance functions are shown in [Figure 9](#). Here we see a combination of encoding parameters (time-independent), and decay parameters for retinotopic stimulus memories (multiplied by τ), fixation locations (multiplied by t) and saccade vectors (multiplied by $t + T$). The reason for multiplying saccade decay constants by the sum of t and T is that a reconstructed fixation location based upon saccade vectors will have a cumulative increase in uncertainty with each saccade since that fixation.

Finally, the likelihood for the interference models is the weighted sum of two likelihood ($e^{\mathcal{L}(k)}$) functions of the form above, but with one computed from the target location and the other from the distractor location. The interference parameter determined the weighting towards the distractor.

We used relatively conservative priors, which were (for all parameters) normal distributions centred on zero, with a spherical covariance of 1/8. This ensured we needed good evidence for the parameters to be estimated as being confidently non-zero. All parameters were log scaling parameters (ensuring the scaling parameters were positive) except for the interference parameter, when present, which was passed through a sigmoid function to constrain the effect of the parameter to lie between zero and one.

Variational Laplace provides approximate Bayesian model evidence values, and posterior probability distributions for each parameter, for each model and participant. By accumulating the log model evidence across all participants and applying a softmax function, we arrived at posterior probabilities for each model. We selected the winning model for subsequent steps of the analysis. Our next stage was to use Parametric Empirical Bayes (PEB)⁷⁶ to assess the effect of diagnostic group on the individual parameter estimates. PEB is effectively a second level linear model. We used the SPM25 implementation `spm_dcm_peb.m`.⁷⁴

Finally, to test hypotheses about the mechanisms that explain performance on complex figure drawing, we took the maximum a posteriori estimates (MAP) for all parameters in the winning model for all participants and used these, normalised by their variance, as regressors for a linear model to predict mean-centred copy scores (omitting those scores > 2 SD from the mean) and changes in score on immediate recall. Again, we used `spm_nlsi_Newton.m` to invert this linear model. Following inversion, we computed the predicted scores based upon the MAP values of the regression coefficients and assessed the percentage of empirical variance explained. The percentage of empirical variance explained by the model was assessed using the formula: $100 \times (1 - \text{var}(\text{residuals}) / \text{var}(\text{empirical_scores}))$.

To assess the robustness of our computational framework and the discriminability of our model space, we performed a post-hoc model recovery analysis. Using synthetic data generated from our candidate models, we confirmed that our inversion scheme could successfully adjudicate between competing hypotheses and that the parameters recovered were informed by the data, moving significantly from their prior expectations. A detailed description of the recovery analysis, including the confusion matrix and parameter correlation coefficients, is provided in Supplementary Materials “Model validation and recovery analysis” and Supplementary Figure 3.

The computational models were fit to the trial-by-trial response data using all 160 trials for each participant. We adopted this inclusive approach because our Bayesian framework explicitly estimates precision (random noise) as a distinct parameter from the saccadic updating cost. This allows the model to partition the total variance, ensuring that even if a specific dataset or clinical cohort exhibits higher overall noise, this is captured by the precision parameter rather than incorrectly inflating the estimates for saccade-driven memory impairment. However, we acknowledge that the reduced number of compliant trials in the PD cohort (averaging approximately 63%) represents a limitation for across-cohort comparisons. Despite this attrition, the absolute number of compliant trials in the PD group (mean approximately 25 per condition) remained sufficient for robust parameter estimation and valid statistical inference.

Statistical analysis

All statistical analyses were conducted in MATLAB R2025a and JASP.⁷⁷

For colour memory, we calculated the absolute angular error, defined as the shortest distance on the colour wheel between the target and the reported colour (range 0 to π radians). For the primary statistical analyses, we utilised the mean absolute error per condition for each participant. By analysing these condition-wise means rather than trial-wise raw data, we invoke the Central Limit Theorem, which ensures that the sampling distribution of these means approximates normality. Because the absolute errors in this paradigm were relatively small and did not approach the π boundary (Figure 5B⁷⁷) even in the clinical cohorts, the data were treated as a continuous measure in our linear ANOVAs and regression models. Moreover, because location and colour recall involve different scales and units, all analyses were performed independently for each feature to avoid cross-dimensional magnitude comparisons.

For correlations, Spearman's rho was reported with two-tailed p-values ($\alpha = 0.05$) and effect size (Fisher's z). Bayesian comparisons were also performed, with Bayes Factor 10 (BF_{10}) reported when applicable. For ANOVAs, when Mauchly's test of sphericity indicated that the assumption of sphericity was violated ($p < 0.05$), and Greenhouse-Geisser correction was used when sphericity was violated. F-stats, p values, and partial eta-squared were reported. Note that sphericity correction is not available for factors with two levels, such as the item factor (only two items). For multiple comparisons, Bonferroni-corrected p-values ($p(\text{bonf})$) were reported.

To quantify the evidence for the findings, particularly regarding null effects in the colour domain, we supplemented our frequentist analyses with Bayesian repeated measures ANOVAs. We calculated inclusion Bayes factors (BF_{incl}) to determine the evidence for including specific factors across the model space, and BF_{01} to quantify evidence in favour of the null hypothesis. Bayes factors were interpreted using the classification scheme, where $BF > 3$ is considered moderate evidence and $BF > 100$ is considered decisive evidence. All Bayesian analyses were conducted in JASP using default prior distributions.

Multiple linear regression models were fit for each participant using all trials. The regression analyses were conducted via the MATLAB function *regress*. The resulting beta coefficients were then averaged across participants. A one-sample t-test (two-tailed) was used to determine if the group average beta coefficient for each predictor was significantly different from zero. The t-statistic and p-value (two-tailed) are reported for each predictor.

Mediation analysis with multiple mediators was conducted in JASP, using standardised regression coefficients⁷⁷. The bias-corrected bootstrap method with 1000 iterations was used to estimate confidence intervals for the indirect and direct effects.⁷⁸

The bayes factor was computed as the division of accumulated posterior probability between the winning model and the next best alternative.

For comparing group differences in the estimated parameters of the winning model, a parametric empirical Bayes (PEB) approach was utilised, employing *spm_dcm_peb* in SPM25. This hierarchical Bayesian method allows for robust group-level inference by modelling individual parameter estimates as samples drawn from a group distribution, thereby accounting for inter-subject variability and propagating uncertainty from the first (individual) level. For each parameter of interest, the group-level posterior mean and covariance were obtained from the PEB analysis. To assess statistically significant differences between groups for a given parameter, the probability of a directional difference between the posterior estimates of any two groups was calculated using their respective posterior distributions. A difference was deemed statistically significant if this calculated probability was greater than 0.95 (corresponding to strong evidence in favour of a directional difference).

Eyetracking recording

Participants sat in front of a 21" CRT computer screen (1024 × 768 pixels; 100 Hz refresh rate) at a viewing distance of 60 cm in a dimly lit quiet room with their head supported on a chinrest. Visual stimuli were presented using MATLAB (The MathWorks) and Psychophysics Toolbox ^{79,80} on a Windows-XP computer. Eye positions during the LOCUS task were monitored using a frame-mounted infrared eye-tracking camera (Eyelink 1000 Tower Mounted, SR Research Ltd.). Right eye was recorded at a sampling rate of 1000 Hz.

Prior to the experiment, a standard nine-point calibration and validation procedure was performed. Participants were instructed to fixate a small black circle with a white centre (0.5 degrees) as it appeared sequentially at nine points forming a 3 × 3 grid across the screen. Calibration was accepted only if the mean validation error was below 0.5 degrees and the maximum error at any single point was below 1.0 degree. If these criteria were not met, or if the experimenter noticed significant gaze drift between blocks, the calibration procedure was repeated. This calibration ensured high spatial accuracy across the entire display area, facilitating the precise monitoring of fixations on item frames and saccadic movements to the response colour wheel.

Eyetracking data analysis

All eyetracking analysis was performed offline. Interval where the eye tracker detected full and partial eye closures were automatically treated as missing data. Furthermore, intervals where pupil diameter was recorded as 0 or change dramatically immediately pre- or post-blink were also automatically treated as blinks and removed. Gaze positions were epoched relative to stimulus onset (item 1, item 2 and colour wheel) and analysed for 1 second post-stimulus-onset. Euclidean distance that the gaze travelled to the target was computed and converted to visual degrees. Fixations were defined as gaze remaining within 2.5° of the target for at least 150 ms. Saccades were defined as eye movements exceeding 150° visual angle per second, lasting longer than 10 ms in duration, and traversing a distance greater than 9.2° from the initial fixation. The distance of 9.2° was defined as it is the radius of perifoveal visual field. ^{81,82}

Task compliance was assessed based on fixation and saccade criteria specific to each condition. For example, in the Saccade-After-Item-1 condition, compliance required fixations on both Item 1 and Item 2, with no saccade exceeding 9.2° after Item 2 measured in the epoch after the onset of colour wheel.

Eye-tracking analysis confirmed high compliance overall, with participants correctly maintaining fixation or executing saccades on the vast majority of trials (83% across all participants). A mixed ANOVA revealed a main effect of group on trial retention ($F(3, 80) = 8.06, p < 0.001$, partial $\eta^2 = 0.23$), primarily due to lower compliance in the PD cohort (YC: 97±4%; EC: 91±10%; AD: 95±5%; PD: 63±38%). Importantly, there was no significant interaction between group and saccade condition ($F(3.36, 80) = 1.78, p = 0.15$, partial $\eta^2 = 0.008$), suggesting that trial attrition was not disproportionately affected by specific task demands in any group.

We acknowledge that this reduced trial count in the PD group represents a limitation for across-cohort comparison. However, the absolute number of compliant trials in PD group (mean approx. 25 per condition) remained sufficient for robust trial-by-trial parameter estimation. Furthermore, the lack of a significant group-by-condition interaction confirms that the results reported for this cohort remain valid and that our primary finding of a selective spatial memory deficit is robust to these differences in data retention.

Data availability

All experimental data and healthy participants' Rey-Osterrieth Complex Figure drawings are publicly available on the Open Science Framework (OSF) at <https://osf.io/95ecp/> . All scripts used for the computational modelling are included in the same repository.

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Additional information

Author Contributions

S.Z.: Conceptualization, Methodology, Formal analysis, Data Curation, Writing - Original Draft, Writing – Review & Editing, Visualization, Project administration. **T.P.:** Methodology, Formal analysis, Writing - Original Draft, Writing – Review & Editing, Visualization. **R.U.:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Project administration. **V.K.:** Methodology, Resources, Investigation, Data Curation, Writing - Review & Editing, Project administration. **G.D.J.:** Resources, Data Curation. **A.S.:** Data Curation. **S.T.:** Conceptualization, Resources, Data Curation, Writing - Review & Editing. **S.M.:** Conceptualization, Methodology, Writing – Review & Editing, Supervision. **M.H.:** Conceptualization, Methodology, Resources, Writing – Review & Editing, Project administration, Supervision, Funding acquisition.

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NIHR Oxford Health BRC		Masud Husain
NIHR NIHR Oxford Biomedical Research Centre (OxBRC)		Sanjay G Manohar

Author ORCID iDs

Sijia Zhao: <https://orcid.org/0000-0002-6246-0702>

Thomas Parr: <https://orcid.org/0000-0001-5108-5743>

Rob Udale: <https://orcid.org/0000-0003-1704-554X>

Verena Klar: <https://orcid.org/0000-0003-1901-2485>

Sofia Toniolo: <https://orcid.org/0000-0002-1833-4995>

Sanjay G Manohar: <https://orcid.org/0000-0003-0735-4349>

Masud Husain: <https://orcid.org/0000-0002-6850-9255>

Additional files

[Supplementary Materials](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study employed a saccade-shifting sequential working memory paradigm, manipulating whether a saccade occurred after each memory array to directly compare retinotopic and transsaccadic working memory for both spatial location and color. Across four participant groups (young and older healthy adults, and patients with Parkinson's disease and Alzheimer's disease), the authors found a consistent saccade-related cost specifically for spatial memory - but not for color - regardless of differences in memory precision. Using computational modeling, they demonstrate that data from healthy participants are best explained by a complex saccade-based updating model that incorporates distractor

interference. Applying this model to the patient groups further elucidates the sources of spatial memory deficits in PD and AD. The authors then extend the model to explain copying deficits in these patient groups, providing evidence for the ecological validity of the proposed saccade-updating retinotopic mechanism.

Strengths:

Overall, the manuscript is well written, and the experimental design is both novel and appropriate for addressing the authors' key research questions. I found the study to be particularly comprehensive: it first characterizes saccade-related costs in healthy young adults, then replicates these findings in healthy older adults, demonstrating how this "remapping" cost in spatial working memory is age-independent. After establishing and validating the best-fitting model using data from both healthy groups, the authors apply this model to clinical populations to identify potential mechanisms underlying their spatial memory impairments. The computational modeling results offer a clearer framework for interpreting ambiguities between allocentric and retinotopic spatial representations, providing valuable insight into how the brain represents and updates visual information across saccades. Moreover, the findings from the older adult and patient groups highlight factors that may contribute to spatial working memory deficits in aging and neurological disease, underscoring the broader translational significance of this work.

Comments on revisions:

The authors have addressed my earlier concerns.

<https://doi.org/10.7554/eLife.109581.2.sa3>

Reviewer #2 (Public review):

Summary:

Zhao et al investigate how object location and colour are degraded across saccadic eye movements. They employ an eye-tracking task that requires participants to remember two sequentially presented items and subsequently report the colour and position of either one of these. Through counterbalancing of the presence or absence of saccades across items, the authors endeavour to dissect the impact of saccades independently on item location or colour. These behavioural findings form the basis of generative models designed to test competing, nested accounts of how stored information is stored and updated across saccades.

Strengths:

The combination of eye-tracking and generative modelling is a strength of the paper, which opens new perspectives into the impact of Alzheimer's and Parkinson's disease on the performance of visuospatial cognitive tests. The finding that the model parameters covary with clinical performance on the ROCF test is a nice example of a "computational assay" of disease.

Comments on revisions:

I thank the authors for their detailed responses and revisions arising from my feedback on the original manuscript. The revised manuscript adequately addresses all of my concerns.

<https://doi.org/10.7554/eLife.109581.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript introduces a visual paradigm aimed at studying tran-saccadic memory.

The authors observe how memory of object location is selectively impaired across eye movements, whereas object colour memory is relatively immune to intervening eye movements.

Results are reported for young and elderly healthy controls, as well as PD and AD participants.

A computational model is introduced to account for these results, indicating how early differences in memory encoding and decay (but not tran-saccadic updating per se) can account for the observed differences between healthy controls and clinical groups.

In the revised manuscript, the authors have addressed most of my initial concerns. The dataset is generally compelling, as it includes healthy younger and older adults as well as clinical populations. In addition, the authors propose an interesting modelling approach designed to isolate and characterize the key components underlying the observed patterns of results.

It is important to acknowledge potential limitations of the modelling approach, particularly the differences in the number of parameters across the tested models. As models with more parameters typically achieve better fit, this issue warrants careful consideration. The authors have substantially addressed this point in their rebuttal.

Concerns regarding the specificity of the findings were also raised and have been adequately discussed in the authors' response. Specifically, they clarified the selective impact of saccade-related costs on spatial working memory updating across eye movements-without affecting feature-based memory (e.g., color) -as well as the specificity of the updating effects observed with the Rey-Osterrieth Complex Figure.

<https://doi.org/10.7554/eLife.109581.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Thank you so much for your comprehensive and insightful assessment of our manuscript. We appreciate your recognition of the novelty of our experimental design and the utility of our computational framework for interpreting visual remapping across the lifespan and in clinical populations. We are very grateful for your suggestions regarding the narrative flow, which have helped us to improve the manuscript's focus and coherence. Our responses to your specific concerns are detailed below.

(1) Relevance of the figure-copy results (pp. 13-15). Is it necessary to include the figure-copy task results within the main text? The manuscript already presents a clear and coherent narrative without this section. The figure-copy task represents a substantial shift from the LOCUS paradigm to an entirely different task that does not measure the same construct. Moreover, the ROCF findings are not fully consistent with the LOCUS results, which introduces confusion and weakens the manuscript's coherence. While I understand the authors' intention to assess the ecological validity of their model, this section does not effectively strengthen the manuscript and may be better removed or placed in the Supplementary Materials.

We thank the reviewer for their perspective regarding the narrative flow and the transition between the LOCUS paradigm and the ROCF results. However, we remain keen to retain these

findings in the main text, as they provide critical ecological and clinical validation for the computational mechanisms identified in our study.

We think these results strengthen the manuscript for the following main reasons:

- (1) The ROCF we used is a standard neuropsychological tool for identifying constructional apraxia. Our results bridge the gap between basic cognitive neuroscience and clinical application by demonstrating that specific remapping parameters—rather than general memory precision—predict real-world deficits in patients.
- (2) The finding that our winning model explains approximately 62% of the variance in ROCF copy scores across all diagnostic groups further indicates that these parameters from the LOCUS task represent core computational phenotypes that underpin complex, real-life visuospatial construction (copying drawings).
- (3) Previous research has often observed only a weak or indirect link between drawing ability and traditional working memory measures, such as digit span (Senese et al., 2020). This was previously attributed to “deictic” strategies—like frequent eye and hand movements—that minimise the need to hold large amounts of information in memory (Ballard et al., 1995; Cohen, 2005; Draschkow et al., 2021). While our study was not exclusively designed to catalogue all cognitive contributions to drawing, the findings provide significant and novel evidence indicating that transsaccadic integration is a critical driver of constructional (copying drawing) ability. By demonstrating this link, the results provide evidence to stimulate a new direction for future research, shifting the focus from general memory capacity toward the precision of spatial updating across eye movements.

In summary, by including the ROCF results in the main text, we provide evidence for a functional role for spatial remapping that extends beyond perceptual stability into the domain of complex visuomotor control. We have expanded on these points throughout the revised manuscript:

In the Introduction: p.2:

“The clinical relevance of these spatial mechanisms is underscored by significant disruptions to visuospatial processing and constructional apraxia—a deficit in copying and drawing figures—observed in neurodegenerative conditions such as Alzheimer’s disease (AD) and Parkinson’s disease (PD).[20,21] This raises a crucial question: do clinical impairments in complex visuomotor tasks stem from specific failures in transsaccadic remapping? If so, the computational parameters that define normal spatial updating should also provide a mechanistic account of these clinical deficits, differentiating them from general age-related decline.”

p.3: “Finally, by linking these mechanistic parameters to a standard clinical measure of constructional ability (the Rey-Osterrieth Complex Figure task), we demonstrate that transsaccadic updating represents a core computational phenotype underpinning real-world visuospatial construction in both health and neurodegeneration.

In the Results:

“To assess whether the mechanistic parameters derived from the LOCUS task represent core phenotypes of real-world visuospatial abilities, we also instructed all participants to complete the Rey-Osterrieth Complex Figure copy task (ROCF; Figure 7A) on an Android tablet using a digital pen (see examples in Figure 7B; all Copy data are available in the open dataset: <https://osf.io/95ecp/>). The ROCF is a gold-standard neuropsychological tool for identifying constructional apraxia.[29] Historically, drawing performance has shown only weak or indirect correlations with traditional working memory measures.[30] This disconnect has been attributed to active visual-sampling strategies—frequent eye movements that treat the

environment as an external memory buffer, minimising the necessity of holding large volumes of information in internal working memory.[3–5]

We hypothesised that drawing accuracy is primarily constrained by the precision of spatial updating across frequent saccades rather than raw memory capacity. To evaluate the ecological validity of the identified saccade-updating mechanism, we modelled individual ROCF copy scores across all four groups using the estimated (maximum a posteriori) parameters from the winning “Dual (Saccade) + Interference” model (Model 7; Figure 8) as regressors in a Bayesian linear model. Prior to inclusion, each regressor was normalised by dividing by the square root of its variance.

This model successfully explained 61.99% of the variance in ROCF copy scores, indicating that these computational parameters are strong predictors of real-world constructional ability (Figure 8A). ... This highlights the critical role of accurate remapping based on saccadic information; even if the core saccadic update mechanism is preserved across groups (as shown in previous analyses), the precision of this updating process is crucial for complex visuospatial tasks. Moreover, worse ROCF copy performance is associated particularly with higher initial angular encoding error. This indicates that imprecision in the initial registration of angular spatial information contributes to difficulties in accurately reproducing complex visual stimuli.”

In the Discussion:

“Importantly, our computational framework establishes a direct mechanistic link between transsaccadic updating and real-world constructional ability. Specifically, higher saccade and angular encoding errors contribute to poorer ROCF copy scores. By mapping these mechanistic estimates onto clinical scores, we found that the parameters derived from our winning model explain approximately 62% of the variance in constructional performance across groups. These findings suggest that the computational parameters identified in the LOCUS task represent core phenotypes of visuospatial ability, providing a mechanistic bridge between basic cognitive theory and clinical presentation.

This relationship provides novel insights into the cognitive processes underlying drawing, specifically highlighting the role of transsaccadic working memory. Previous research has primarily focused on the roles of fine motor control and eye-hand coordination in this skill. [4,50–55] This is partly because of consistent failure to find a strong relation between traditional memory measures and copying ability [4,31] For instance, common measures of working memory, such as digit span and Corsi block tasks, do not directly predict ROCF copying performance.[31,56] Furthermore, in patients with constructional apraxia, these memory performance measures often remain relatively preserved despite significant drawing impairments.[56–58] In the literature, this lack of association has often been attributed to “deictic” visual-sampling strategies, characterised by frequent eye movements that treat the environment as an external memory buffer, thereby minimising the need to maintain a detailed internal representation.[4,59] In a real-world copying task, the ROCF requires a high volume of saccades, making it uniquely sensitive to the precision of the dynamic remapping signals identified here. Recent eye-tracking evidence confirms that patients with AD exhibit significantly more saccades and longer fixations during figure copying compared to controls, potentially as a compensatory response to transsaccadic working memory constraints.[56] This high-frequency sampling—averaging between 150 and 260 saccades for AD patients compared to approximately 100 for healthy controls—renders the task highly dependent on the precision of dynamic remapping signals.[56] To ensure this relationship was not driven by a general “g-factor” or non-spatial memory impairment, we further investigated the role of broader cognitive performance using the ACE-III Memory subscale. We found that the relationship between transsaccadic working memory and ROCF performance remains highly significant, even after controlling for age, education, and ACE-III Memory subscore. This suggests that transsaccadic updating may represent a discrete

computational phenotype required for visuomotor control, rather than a non-specific proxy for global cognitive decline.

In other words, even when visual information is readily available in the world, the act of copying depends critically on working memory across saccades. This reveals a fundamental computational trade-off: while active sampling strategies (characterised with frequent eye-hand movements) effectively reduce the load on capacity-limited working memory, they simultaneously increase the demand for precise spatial updating across eye movements. By treating the external world as an "outside" memory buffer, the brain minimises the volume of information it must hold internally, but it becomes entirely dependent on the reliability with which that information is remapped after each eye movement. This perspective aligns with, rather contradicts, the traditional view of active sampling, which posits that individuals adapt their gaze and memory strategies based on specific task demands.[3,60] Furthermore, this perspective provides a mechanistic framework for understanding constructional apraxia; in these clinical populations, the impairment may not lie in a reduced memory "span," but rather in the cumulative noise introduced by the constant spatial remapping required during the copying process.[58,61]

Beyond constructional ability, these findings suggest that the primary evolutionary utility of high-resolution spatial remapping lies in the service of action rather than perception. While spatial remapping is often invoked to explain perceptual stability,[11–13,15] the necessity of high-resolution transsaccadic memory for basic visual perception is debated.[13,62–64] A prevailing view suggests that detailed internal models are unnecessary for perception, given the continuous availability of visual information in the external world.[13,44] Our findings support an alternative perspective, aligning with the proposal that high-resolution transsaccadic memory primarily serves action rather than perception.[13] This is consistent with the need for precise localisation in eye-hand coordination tasks such as pointing or grasping.[65] Even when unaware of intrasaccadic target displacements, individuals rapidly adjust their reaching movements, suggesting direct access of the motor system to remapping signals.[66] Further support comes from evidence that pointing to remembered locations is biased by changes in eye position,[67] and that remapping neurons reside within the dorsal "action" visual pathway, rather than the ventral "perception" visual pathway.[13,68,69] By demonstrating a strong link between transsaccadic working memory and drawing (a complex fine motor skill), our findings suggest that precise visual working memory across eye movements plays an important role in complex fine motor control."

(2) Model fitting across age groups (p. 9).

It is unclear whether it is appropriate to fit healthy young and healthy elderly participants' data to the same model simultaneously. If the goal of the model fitting is to account for behavioral performance across all conditions, combining these groups may be problematic, as the groups differ significantly in overall performance despite showing similar remapping costs. This suggests that model performance might differ meaningfully between age groups. For example, in Figure 4A, participants 22-42 (presumably the elderly group) show the best fit for the Dual (Saccade) model, implying that the Interference component may contribute less to explaining elderly performance.

Furthermore, although the most complex model emerges as the best-fitting model, the manuscript should explain how model complexity is penalized or balanced in the model comparison procedure. Additionally, are Fixation Decay and Saccade Update necessarily alternative mechanisms? Could both contribute simultaneously to spatial memory representation? A model that includes both mechanisms-e.g., Dual (Fixation) + Dual (Saccade) + Interference-could be tested to determine whether it outperforms Model 7 to rule out the sole contribution of complexity.

We thank you for the opportunity to expand upon and clarify our modelling approach. Our decision to use a common generative model for both young and older adults was grounded in the empirical finding that there was no significant interaction between age group and saccade condition for either location or colour memory. While older adults demonstrated lower baseline precision, the specific "saccade cost" remained remarkably consistent across cohorts. This was the justification we proceeded on to use of a common model to assess quantitative differences in parameter estimates while maintaining a consistent mechanistic framework for comparison.

Moreover, our winning model nests simpler models as special cases, providing the flexibility to naturally accommodate groups where certain components—such as interference—might play a reduced role. This ultimately confirms that the mechanisms for age-related memory deficits in this task reflect more general decline rather than a qualitative failure of the saccadic remapping process.

This approach is further supported by the properties of the Bayesian model selection (BMS) procedure we used, which inherently penalises the inclusion of unnecessary parameters. Unlike maximum likelihood methods, BMS compares marginal likelihoods, representing the evidence for a model integrated over its entire parameter space. This follows the principle of Bayesian Occam's Razor, where a model is only favoured if the improvement in fit justifies the additional parameter space; redundant parameters instead "dilute" the probability mass and lower the model evidence.

Consequently, we contend that a hybrid model combining fixation and saccade mechanisms is unnecessary, as we have already adjudicated between alternative mechanisms of equal complexity. Specifically, Model 6 (Dual Fixation + Interference) and Model 7 (Dual Saccade + Interference) possess an identical number of parameters. The fact that Model 7 emerged as the clear winner—providing substantial evidence against Model 6 with a Bayes Factor of 6.11—demonstrates that our model selection is driven by the specific mechanistic account of the data rather than a simple preference for complexity.

We have revised the Results and Discussion sections of the manuscript to state these points more explicitly for readers and have included references to established literature regarding the robustness of marginal likelihoods in guarding against overfitting.

In the Results,

"By fitting these models to the trial-by-trial response data from all healthy participants (N=42), we adjudicated between competing mechanisms to determine which best explained participant performance (Figure 4). We used random-effects Bayesian model selection to identify the most plausible generative model. This process relies on the marginal likelihood (model evidence), which inherently balances model fit against complexity—a principle often referred to as Occam's razor.[25–27] The analysis yielded a strong result: the "Dual (Saccade) + Interference" model (Model 7 in Table 1) emerged as the winning model, providing substantial evidence against the next best alternative with a Bayes Factor of 6.11."

In the Discussion:

"Our framework employs Variational Laplace, a method used to recover computational phenotypes in clinical populations like those with substance use disorders,[34,35] and the models we fit using this procedure feature time-dependent parameterisation of variance—conceptually similar to the widely-used Hierarchical Gaussian Filter.[36–39] Importantly, the risk of overfitting is mitigated by the Bayesian Model Selection framework; by utilising the marginal likelihood for model comparison, the procedure inherently penalises excessive model complexity and promotes generalisability.[25–27,40] This generalisability was further evidenced by the model's ability to predict performance on the independent ROCF task,

confirming that these parameters represent robust mechanistic phenotypes rather than idiosyncratic fits to the initial dataset.”

Minor point: On p. 9, line 336, Figure 4A does not appear to include the red dashed vertical line that is mentioned as separating the age groups.

Thank you for pointing out this inconsistency. We apologise for the oversight; upon further review, we concluded that the red dashed vertical line was unnecessary for the clear presentation of the data. We have therefore removed the line from Figure 4A and deleted the corresponding sentence in the figure caption.

(3) Clarification of conceptual terminology.

Some conceptual distinctions are unclear. For example, the relationship between "retinal memory" and "transsaccadic memory," as well as between "allocentric map" and "retinotopic representation," is not fully explained. Are these constructs related or distinct? Additionally, the manuscript uses terms such as "allocentric map," "retinotopic representation," and "reference frame" interchangeably, which creates ambiguity. It would be helpful for the authors to clarify the relationships among these terms and apply them consistently.

Thank you for pointing this out. We have revised the manuscript to ensure that these terms are applied with greater precision and consistency. Our revisions standardise the terminology based on the following distinctions:

Reference frames: We distinguish between the eye-centred reference frame (coordinate systems that shift with gaze) and the world-centred reference frame (coordinate systems anchored to the environment).

Retinotopic representation vs. allocentric map: We clarify that retinotopic representations are encoded within an eye-centred reference frame and are updated with every ocular movement. Conversely, the allocentric map is anchored to stable environmental features, remaining invariant to the observer’s gaze direction or position.

Retinotopic memory vs. transsaccadic memory: We have removed the term "retinal memory" to avoid ambiguity. We now consistently use retinotopic memory to describe the persistence of visual information in eye-centred coordinates within a single fixation. In contrast, transsaccadic memory refers to the higher-level integration of visual information across saccades, which involves the active updating or remapping of representations to maintain stability.

To incorporate these clarifications, we have implemented the following changes:

In the Introduction, the second paragraph has been entirely rewritten to establish these definitions at the outset, providing a clearer theoretical framework for the study.

“Central to this enquiry is the nature of the coordinate system used for the brain’s internal spatial representation. Does the brain maintain a single, world-centred (allocentric) map, or does it rely on a dynamic, eye-centred (retinotopic) representation?[11,13,15,16] In the latter system, retinotopic memory preserves spatial information within a fixation, whereas transsaccadic memory describes the active process of updating these representations across eye movements to achieve spatiotopic stability—the perception of a stable world despite eye movements.[11,16–18] If spatial stability is indeed reconstructed through such remapping, the mechanism remains unresolved: do we retain memories of absolute fixation locations, or do we reconstruct these positions from noisy memories of the intervening saccade vectors? We can test these hypotheses by analysing when and where memory errors occur. Assuming

that memory precision declines over time,[19] the resulting error distributions should reveal the specific variables that are represented and updated across each saccade.”

In the Results, the opening section of the Results has been reorganised to align with this terminology. We have ensured that the hypotheses and behavioural data—specifically the definition of “saccade cost”—are introduced using this consistent conceptual vocabulary to improve the overall coherence of the narrative.

(4) Rationale for the selective disruption hypothesis (p. 4, lines 153-154). The authors hypothesize that “saccades would selectively disrupt location memory while leaving colour memory intact.” Providing theoretical or empirical justification for this prediction would strengthen the argument.

We have revised the Results to state the hypothesis more explicitly and expanded the Discussion to provide a robust theoretical and empirical rationale:

In the Results,

“This design allowed us to isolate and quantify the unique impact of saccades on spatial memory, enabling us to test competing hypotheses regarding spatial representation. If spatial memory were solely underpinned by an allocentric mechanism, precision should remain comparable across all conditions as the representation would be world-centred and unaffected by eye movements. Thus, performance in the no-saccade condition should be comparable to the two-saccade condition. Conversely, if spatial memory relies on a retinotopic representation requiring active updating across eye movements, the two-saccade condition was anticipated to be the most challenging due to cumulative decay in the memory traces used for stimulus reconstruction after each saccade.[22] Critically, we hypothesised that this saccade cost would be specific to the spatial domain; while location requires active remapping via noisy oculomotor signals, non-spatial features like colour are not inherently tied to coordinate transformations and should therefore remain stable (see more in Discussion below).

Meanwhile, the no-saccade condition was expected to yield the most accurate localisation, relying solely on retinotopic information (retinotopic working memory). These predictions were confirmed in young healthy adults ($N = 21$, mean age = 24.1 years, ranged between 19 and 34). A repeated measures ANOVA revealed a significant main effect of saccades on location memory ($F(2.2, 43.9) = 33.2$, $p < 0.001$, partial $\eta^2 = 0.62$), indicating substantial impairment after eye movements (Figure 2A). In contrast, colour memory remained remarkably stable across all saccade conditions (Figure 2B; $F(2.2, 44.7) = 0.68$, $p = 0.53$, partial $\eta^2 = 0.03$).

This “saccade cost”—the loss of memory precision following an eye movement—indicates that spatial representations require active updating across saccades rather than being maintained in a static, world-centred reference frame.

Critically, our comparison between spatial and colour memory does not rely on the absolute magnitude of errors, which are measured in different units (degrees of visual angle vs. radians). Instead, we assessed the relative impact of the same saccadic demand on each feature within the same trial. While location recall showed a robust saccade cost, colour recall remained statistically unchanged. To ensure this null effect was not due to a lack of measurement sensitivity, we examined the recency effect; recall performance for the second item was predicted to be better than for the first stimulus in each condition.[23,24] As expected, colour memory for Item 2 was significantly more accurate than for Item 1 ($F(1, 20) = 6.52$, $p = 0.02$, partial $\eta^2 = 0.25$), demonstrating that the task was sufficiently sensitive to detect standard working memory fluctuations despite the absence of a saccade-induced deficit.”

In the Discussion, we now write that on p.18:

“A clear finding was the specificity of the saccade cost to spatial features; it was not observed for non-spatial features like colour, even in neurodegenerative conditions. This discrepancy challenges notions of fixed visual working memory capacity unaffected by saccades.^{16,44–46} The differential impact on spatial versus non-spatial features in transsaccadic memory aligns with the established “what” and “where” pathways in visual processing.^{32,33} For objects to remain unified, object features must be bound to stable representations of location across saccades.¹⁹ One possibility is that remapping updates both features and location through a shared mechanism, predicting equal saccadic interference for both colour and location in the present study.

However, our findings suggest otherwise. One potential concern is whether this dissociation simply reflects the inherent spatial noise introduced by fixational eye movements (FEMs), such as microsaccades and drifts.⁴⁷ Because locations are stored in a retinotopic frame, fixational instability necessarily shifts retinal coordinates over time. However, the “saccade cost” here was defined as the error increase relative to a no-saccade baseline of equal duration; because both conditions are subject to the same fixational drift, any FEM-induced noise is effectively subtracted out. Thus, despite the ballistic and non-Gaussian nature of FEMs,⁴⁸ they cannot account for the fact the saccade cost in the spatial memory, but total absence in the colour domain. Another possibility is that this dissociation reflects differences in baseline task difficulty or dynamic range. Yet, the presence of a robust recency effect in colour memory (Figure 2B) confirms that our paradigm was sensitive to memory-dependent variance and was not limited by floor or ceiling effects.

The fact that identical eye movements—executed simultaneously and with identical vectors—systematically degraded spatial precision while sparing colour suggests a feature-specific susceptibility to transsaccadic remapping. This supports the view that the computational process of updating an object’s location involves a vector-subtraction mechanism—incorporating noisy oculomotor commands (efference copies)—that introduces specific spatial variance. Because this remapping is a coordinate transformation, the resulting sensorimotor noise does not functionally propagate to non-spatial feature representations. Consequently, features like colour may be preserved or automatically remapped without the precision loss associated with spatial updating.^{11,49} Our paradigm thus provides a refined tool to investigate the architecture of transsaccadic working memory across distinct object features.”

(5) Relationship between saccade cost and individual memory performance (p. 4, last paragraph).

The authors report that larger saccades were associated with greater spatial memory disruption. It would be informative to examine whether individual differences in the magnitude of saccade cost correlate with participants' overall/baseline memory performance (e.g. their memory precision in the no-saccade condition). Such analyses might offer insights into how memory capacity/ability relates to resilience against saccade-induced updating.

We have now conducted the correlation analysis to determine whether baseline memory capacity (no-saccade condition) predicts resilience to saccade-induced updating. The results indicate that these two factors are independent.

To clarify the nature of the saccade-induced impairment, we have updated the text as follows:

p.4: “This “saccade cost”—the loss of memory precision following an eye movement—indicates that spatial representations require active updating across saccades rather than being maintained in a static, world-centred reference frame.”

p.5: “Further analysis examined whether individual differences in baseline memory precision (no-saccade condition) predicted resilience to saccadic disruption. Crucially, individual saccade costs (defined as the precision loss relative to baseline) did not correlate with baseline precision ($\rho = 0.20$, $p = 0.20$). This suggests that the noise introduced by transsaccadic remapping acts as an independent, additive source of variance that is not modulated by an individual’s underlying memory capacity. These findings imply a functional dissociation between the mechanisms responsible for maintaining a representation and those involved in its coordinate transformation.”

(6) Model fitting for the healthy elderly group to reveal memory-deficit factors (pp. 11-12). The manuscript discusses model-based insights into components that contribute to spatial memory deficits in AD and PD, but does not discuss components that contribute to spatial memory deficits in the healthy elderly group. Given that the EC group also shows impairments in certain parameters, explaining and discussing these outcomes of the EC group could provide additional insights into age-related memory decline, which would strengthen the study’s broader conclusions.

This is a very good point. We rewrote the corresponding results section (p.12-13):

“Modelling reveals the sources of spatial memory deficits in healthy aging and neurodegeneration - To understand the source of the observed deficits, we applied the winning ‘Dual (Saccade) + Interference’ model the data from all participants (YC, EC, AD, and PD). By fitting the model to the entire dataset, we obtained estimates of the parameters for each individual, which then formed the basis for our group-level analysis. To formally test for group differences, we used Parametric Empirical Bayes (PEB), a hierarchical Bayesian approach that compares parameter estimates across groups while accounting for the uncertainty of each estimate [28]. This allowed us to identify which specific cognitive mechanisms, as formalised by the model parameters, were affected by age and disease.

The Bayesian inversion used here allows us to quantify the posterior mode and variance for each parameter and the covariance for each parameter. From these, we can compute the probabilities that pairs of parameters differ from one another, which we report as $P(A > B)$ —meaning the posterior probability that the parameter for group A was greater than that for group B.

We first examined the specific parameters differentiating healthy elderly (EC) from young controls (YC) to isolate the factors contributing to non-pathological, age-related decline. The analysis revealed that healthy ageing is primarily characterised by a significant increase in Radial Decay ($P(EC > YC) = 0.995$), a heightened susceptibility to Interference ($P(EC > YC) = 1.000$), and a reduction in initial Angular Encoding precision ($P(YC < EC) = 0.002$; Figure 6). These results suggest that normal ageing degrades the fidelity of the initial memory trace and its resilience over time, while the core computational process of updating information across saccades remains intact.

Beyond these baseline ageing effects, our clinical cohorts exhibited more severe and condition-dependent impairments. Radial decay showed a clear, graded impairment: AD patients had a greater decay rate than PD patients ($P(AD > PD) = 1.000$), who in turn were more impaired than the EC group ($P(PD > EC) = 0.996$). A similar graded pattern was observed for Interference, where AD patients were most susceptible ($P(AD > PD) = 0.999$), while the PD and EC groups did not significantly differ ($P(PD > EC) = 0.532$).

Patients with AD also showed a tendency towards greater angular decay than controls ($P(AD > EC) = 0.772$), although this fell below the 95% probability threshold. This effect was influenced by a lower decay rate in the PD group compared to the EC group ($P(PD < EC) = 0.037$). In contrast, group differences in encoding were less pronounced. While YC exhibited significantly higher precision than all other groups, AD patients showed significantly higher

angular encoding error than PD patients ($P(AD > PD) = 0.985$), though neither group differed significantly from the EC group.

Crucially, parameters related to the saccade itself—saccade encoding and saccade decay—did not differentiate the groups. This indicates that neither healthy ageing nor the early stages of AD and PD significantly impair the fundamental machinery for transsaccadic remapping. Instead, the visuospatial deficits in these conditions arise from specific mechanistic failures: a faster decay of radial position information and increased susceptibility to interference, both of which are present in healthy ageing but significantly amplified by neurodegeneration.”

In the Discussion, we added:

“Although saccade updating was an essential component of the winning model, its two key parameters—initial encoding error and decay rate during maintenance—did not significantly differ across groups. This indicates that the core computational process of updating spatial information based on eye movements is largely preserved in healthy aging and neurodegeneration.

Instead, group differences were driven by deficits in angular encoding error (precision of initial angle from fixation), angular decay, radial decay (decay in memory of distance from fixation), and interference susceptibility. This implies a functional and neuroanatomical dissociation: while the ventral stream (the “what” pathway) shows an age-related decline in the quality and stability of stored representations, the dorsal-stream (the “where” pathway) parietal-frontal circuits responsible for coordinate transformations remain functionally robust.[31–34] These spatial updating mechanisms appear resilient to the normal ageing trajectory and only break down when challenged by the specific pathological processes seen in Alzheimer’s or Parkinson’s disease.”

(7) Presentation of saccade conditions in Figure 5 (p. 11). In Figure 5, it may be clearer to group the four saccade conditions together within each patient group. Since the main point is that saccadic interference on spatial memory remains robust across patient groups, grouping conditions by patient type rather than intermixing conditions would emphasize this interpretation.

There are several valid ways to present these plots, but we chose this format because it allows for a direct visual comparison of the post-hoc group differences within each specific task demand. This arrangement clearly illustrates the graded impairment from young controls through to patients with Alzheimer’s disease across every condition. This structure also directly mirrors our two-way ANOVA, which identified significant main effects for both Group and Condition, but crucially, no significant Group x Condition interaction. We felt that grouping the data by participant group would force readers to look across four separate clusters to compare the slopes, making the stability of the saccadic remapping mechanism much harder to grasp at a glance.

Reviewer #1 (Recommendations for the authors):

(1) Formatting of statistical parameters.

The formatting of statistical symbols should be consistent throughout the manuscript. Some instances of F , p , and t are italicized, while others are not. All statistical symbols should be italicized.

Thank you for pointing this out. We have audited the manuscript. While we have revised the text to address these instances throughout the Results and Methods sections, any remaining minor formatting inconsistencies will be corrected during the final typesetting stage.

(2) Minor typographical issues.

(a) Line 532: "are" should be "be."

(b) Line 654: "cantered" should be "centered."

(c) Line 213: In " $(p(\text{bonf}) < 0.001, |t| \{ \text{greater than or equal to} \} 5.94)$," the t value should be reported with its degrees of freedom, and t should be reported before p . The same applies to line 215.

Thank you for your careful reading. All corrected.

Reviewer #2 (Public review):

We thank you for your positive feedback regarding our eye-tracking methodology and computational approach. We appreciate your critical insights into the feature-specific disruption hypothesis and the task structure. We have substantially revised the results and discussion about the saccadic interference on colour memory. Below we will answer your suggestions point-by-point:

Reviewer #2 (Recommendations for the authors):

(1) The study treats colour and location errors as comparable when arguing that saccades selectively disrupt spatial but not colour memory. However, these measures are defined in entirely different units (degrees of visual angle vs radians on a colour wheel) and are not psychophysically or statistically calibrated. Baseline task difficulty, noise level, or dynamic range do not appear to be calibrated or matched across features. As a result, the null effect of saccades on colour could reflect lower sensitivity or ceiling effects rather than implicit feature-specific robustness.

We agree that direct comparisons of absolute error magnitudes across different dimensions are not appropriate. Our argument for feature-specific disruption relies not on the scale of errors, but on the presence or absence of a saccade cost within identical trials. In our within-subject design, the same saccade vectors produced a systematic increase in location error while leaving colour error statistically unchanged. To address sensitivity, we observed that colour memory was sufficiently precise to show a significant recency effect ($p = 0.02$). To further quantify the evidence for the null effect, we performed Bayesian repeated measures ANOVAs, which yielded a $\text{BF}_{10} = 0.22$. This provides substantial evidence that saccades do not disrupt colour precision, regardless of baseline sensitivity.

We have substantially revised this in Results, Methods and Discussion:

In the Results:

"This design allowed us to isolate and quantify the unique impact of saccades on spatial memory, enabling us to test competing hypotheses regarding spatial representation. If spatial memory were solely underpinned by an allocentric mechanism, precision should remain comparable across all conditions as the representation would be world-centred and unaffected by eye movements. Thus, performance in the no-saccade condition should be comparable to the two-saccade condition. Conversely, if spatial memory relies on a retinotopic representation requiring active updating across eye movements, the two-saccade condition was anticipated to be the most challenging due to cumulative decay in the memory traces used for stimulus reconstruction after each saccade.[22] Critically, we hypothesised that this saccade cost would be specific to the spatial domain; while location requires active remapping via noisy oculomotor signals, non-spatial features like colour are not inherently tied to coordinate transformations and should therefore remain stable (see more in Discussion below).

Meanwhile, the no-saccade condition was expected to yield the most accurate localisation, relying solely on retinotopic information (retinotopic working memory). These predictions were confirmed in young healthy adults ($N = 21$, mean age = 24.1 years, ranged between 19 and 34). A repeated measures ANOVA revealed a significant main effect of saccades on location memory ($F(2.2,43.9)=33.2$, $p<0.001$, partial $\eta^2=0.62$), indicating substantial impairment after eye movements (Figure 2A). In contrast, colour memory remained remarkably stable across all saccade conditions (Figure 2B; $F(2.2, 44.7) = 0.68$, $p=0.53$, partial $\eta^2 =0.03$).

This “saccade cost”—the loss of memory precision following an eye movement—indicates that spatial representations require active updating across saccades rather than being maintained in a static, world-centred reference frame.

Critically, our comparison between spatial and colour memory does not rely on the absolute magnitude of errors, which are measured in different units (degrees of visual angle vs. radians). Instead, we assessed the relative impact of the same saccadic demand on each feature within the same trial. While location recall showed a robust saccade cost, colour recall remained statistically unchanged. To ensure this null effect was not due to a lack of measurement sensitivity, we examined the recency effect; recall performance for the second item was predicted to be better than for the first stimulus in each condition.[23,24] As expected, colour memory for Item 2 was significantly more accurate than for Item 1 ($F(1,20) = 6.52$, $p = 0.02$, partial $\eta^2 = 0.25$), demonstrating that the task was sufficiently sensitive to detect standard working memory fluctuations despite the absence of a saccade-induced deficit.”

In the Methods, at the beginning of “Statistical Analysis”, we added

“Because location and colour recall involve different scales and units, all analyses were performed independently for each feature to avoid cross-dimensional magnitude comparisons.” (p25)

In the Discussion, we added:

“A potential concern is whether the observed dissociation between colour and location reflects differences in baseline task difficulty or dynamic range. Yet, the presence of a robust recency effect in colour memory (Figure 2B) confirms that our paradigm was sensitive to memory-dependent variance and was not limited by floor or ceiling effects.”

(2) Colour and then location are probed serially, without a counter-balanced order. This fixed response order could introduce a systematic bias because location recall is consistently subject to longer memory retention intervals and cognitive interference from the colour decision. The observed dissociation-saccades impair location but not colour, and may therefore reflect task structure rather than implicit feature-specific differences in trans-saccadic memory.

Thank you for the insightful observation regarding our fixed response order. We acknowledge that that a counterbalanced design is typically preferred to mitigate potential order effects. However, we chose this consistent sequence to ensure the task remained accessible for cognitively impaired patients (i.e., the Alzheimer’s disease (AD) and Parkinson’s disease (PD) cohorts). Conducting an eye-tracking memory task with cognitively impaired patients is challenging, as they may struggle with task engagement or forget complex instructions. During the design phase, we prioritised a consistent structure to reduce the cognitive load and task-switching demands that typically challenge these cohorts.

Critically, because the saccade cost is a relative measure calculated by comparing conditions with identical timings, any bias from the fixed order is present in both the baseline and

saccade trials. The disruption we report is therefore a specific effect of eye movements that goes beyond the noise introduced by the retention interval or the preceding colour report.

We added the following text in the Methods – experimental procedure (p.22):

“Recall was performed in a fixed order, with colour reported before location. This sequence was primarily chosen to minimise cognitive load and task-switching demands for the two neurological patient cohorts, ensuring the paradigm remained accessible for individuals with AD and PD. While this order results in a slightly longer retention interval for location recall, the saccade cost was identified by comparing location error across experimental conditions with similar timings but varying saccadic demands.”

(3) Relatedly, because spatial representations are retinotopic, fixational eye movements (FEMs - microsaccades and drift) displace the retinal coordinates of encoded positions, increasing apparent spatial noise with time delays. Colour memory, however, is feature-based and unaffected by small retinal translations. Thus, any between-condition or between-group differences in FEMs could selectively inflate location error and the associated model parameters (encoding noise, decay, interference), while leaving colour error unchanged. Note that FEMs tend to be slightly ballistic [1,2], hence not well modelled with a Gaussian blur.

This is a very insightful point. We have now addressed this in detail within the discussion:

“However, our findings suggest otherwise. One potential concern is whether this dissociation simply reflects the inherent spatial noise introduced by fixational eye movements (FEMs), such as microsaccades and drifts.[46] Because locations are stored in a retinotopic frame, fixational instability necessarily shifts retinal coordinates over time. However, the “saccade cost” here was defined as the error increase relative to a no-saccade baseline of equal duration; because both conditions are subject to the same fixational drift, any FEM-induced noise is effectively subtracted out. Thus, despite the ballistic and non-Gaussian nature of FEMs, [47] they cannot account for the fact the saccade cost in the spatial memory, but total absence in the colour domain. Another possibility is that this dissociation reflects differences in baseline task difficulty or dynamic range. Yet, the presence of a robust recency effect in colour memory (Figure 2B) confirms that our paradigm was sensitive to memory-dependent variance and was not limited by floor or ceiling effects.”

(4) There is no in silico demonstration that the modelling framework can recover the true generating model from synthetic data or recover accurate parameters under realistic noise levels, which can be challenging in generative models with a hierarchical structure (as per [3], for example). Figure 8b shows that the parameters possess substantial posterior covariance, which raises concerns as to whether they can be reliably disambiguate.

Many thanks for this comment. We have added a simple recovery analysis as detailed below but are also keen to ensure we fully answer your question—which has more to do with empirical rather than simulated data—and make clear the rationale for this analysis in this instance.

We added this in Supplementary Materials:

“Model validation and recovery analysis

The following section provides a detailed technical assessment of the model inversion scheme, focusing on the discriminability of the model space and the identifiability of individual parameters.

Recovery analyses of this sort are typically used prior to collecting data to allow one to determine whether, in principle, the data are useful in disambiguating between hypotheses. In this sense, they have a role analogous to a classical power calculation. However, their utility is limited when used post-hoc when data have already been collected, as the question of whether the models can be disambiguated becomes one of whether non-trivial Bayes factors can be identified from those data.

The reason for including a recovery analysis here is not to identify whether the model inversion scheme identifies a ‘true’ model. The concept of ‘true generative models’ commits to a strong philosophical position which is at odds with the ‘all models are wrong, but some are useful’ perspective held by many in statistics, e.g., (So, 2017). Of note, one can always confound a model recovery scheme by generating the same data in a simple way, and in (one of an infinite number of) more complex ways. A good model inversion scheme will always recover the simple model and therefore would appear to select the ‘wrong’ model in a recovery analysis. However, it is still the best explanation for the data. For these reasons, we do not necessarily expect ‘good’ recoverability in all parameter ranges. This is further confounded by the relationship between the models we have proposed—e.g., an interference model with very low interference will look almost identical to a model with no interference. The important question here is whether they can be disambiguated with real data.

Instead, the value of a post-hoc recovery analysis here is to evaluate whether there was a sensible choice of model space—i.e., that it was not a priori guaranteed that a single model (and, specifically, the model we found to be the best explanation for the data) would explain the results of all others. To address this, for each model, we simulated 16 datasets, each of which relied upon parameters sampled from the model priors, which included examples of each of the experimental conditions. We then fit each of these datasets to each of the 7 models to construct the confusion matrix shown in the lower panel of Supplementary Figure 3, by accumulating evidence over each of the 16 participants generated according to each ‘true’ model (columns) for each of the possible explanatory models (rows). This shows that no one model, for the parameter ranges sampled here, explains all other datasets. Interestingly, our ‘winning’ model in the empirical analysis is not the best explanation for any of the datasets simulated (including its own). This is reassuring, in that it implies this model winning was not a foregone conclusion and is driven by the data—not just the choice of model space.”

Your point about the posterior covariance is well founded. As we describe in Supplementary Materials, this is an inherent feature of inverse problems (analogous to EEG source localisation). However, the fact that our posterior densities move significantly away from the prior expectations demonstrates that the data are indeed informative. By adopting a Bayesian framework, we are able to explicitly quantify this uncertainty rather than ignoring it, providing a more transparent account of parameter identifiability. We have added the following in the same section of Supplementary Materials:

“This problem is an inverse problem—inferring parameters from a non-linear model. We therefore expect a degree of posterior covariance between parameters and, consequently, that they cannot be disambiguated with complete certainty. While some degree of posterior covariance is inherent to inverse models—including established methods like EEG source localisation—the fact that many of the parameters are estimated with posterior densities that do not include their prior expectations implies the data are informative about these.

The advantage of the Bayesian approach we have adopted here is that we can explicitly quantify posterior covariance between these parameters, and therefore the degree to which they can be disambiguated. While the posterior covariance matrices from empirical data are the relevant measure here, we can better understand the behaviour of the model inversion scheme in relation to the specific models used using the model recovery analysis reported in Supplementary figure 3.

The middle panel of the figure is key, along with the correlation coefficients reported in the figure caption. Here, we see at least a weak positive correlation (in some cases much stronger) for almost all parameters and limited movement from prior expectations for those parameters that are less convincingly recovered. This reinforces that the ability of the scheme to recover parameters is best assessed in terms of the degree of movement of posterior from prior values following fitting to empirical data.”

(5) The authors employ Bayes factors (BFs) to disambiguate models, but BFs would also strengthen the claims that location, but not colour, is impacted by saccades. Despite colour being a circular variable, colour error is analysed using ANOVA on linearised differences (radians). The authors should also arguably use circular statistics, such as the von Mises distribution, for the analysis of colour.

Regarding the use of circular statistics, you are correct that such error distributions are not suitable for ANOVA, and it is better to use circular statistics. However, for the present dataset, we used the mean absolute angular error per condition (ranging from 0 to π radians), which represents the shortest distance on the colour wheel between the target and the response.

This approach effectively linearises the measure by removing the 2π wrap-around boundary because the observed errors were relatively small and did not cluster near the π boundary—even in the patient cohorts (Figure 5B)—the “wrap-around” effect of circular space is negligible. Moreover, by analysing the mean error across trials for each condition, rather than trial-wise data, we invoke the Central Limit Theorem. This ensures that the distribution of these means is approximately normal, satisfying the fundamental assumptions of ANOVA. Due to these reasons, we adopted simpler linear models. We confirmed that the data did not violate the assumptions of linear statistics. In this low-noise regime, linear and circular models converge on the same conclusions. This has been revised in Methods:

“For colour memory, we calculated the absolute angular error, defined as the shortest distance on the colour wheel between the target and the reported colour (range 0 to π radians). For the primary statistical analyses, we utilised the mean absolute error per condition for each participant. By analysing these condition-wise means rather than trial-wise raw data, we invoke the Central Limit Theorem, which ensures that the sampling distribution of these means approximates normality. Because the absolute errors in this paradigm were relatively small and did not approach the π boundary (Figure 5B) even in the clinical cohorts, the data were treated as a continuous measure in our linear ANOVAs and regression models. Moreover, because location and colour recall involve different scales and units, all analyses were performed independently for each feature to avoid cross-dimensional magnitude comparisons.”

We have also now integrated Bayesian repeated measures ANOVA throughout the manuscript. The Results section for the young healthy adults now reads (p. 4):

“A repeated measures ANOVA revealed a significant main effect of saccades on location memory ($F(3, 20) = 51.52$, $p < 0.001$, partial $\eta^2 = 0.72$), with Bayesian analysis providing decisive evidence for the inclusion of the saccade factor ($BF_{\text{incl}} = 3.52 \times 10^{13}$, $P(\text{incl} | \text{data}) = 1.00$). In contrast, colour memory remained remarkably stable across all saccade conditions ($F(3, 20) = 0.57$, $p = 0.64$, partial $\eta^2 = 0.03$). This null effect was supported by Bayesian analysis, which provided moderate evidence in favour of the null hypothesis ($BF_{01} = 8.46$, $P(\text{excl} | \text{data}) = 0.89$), indicating that the data were more than eight times more likely under the null model than a model including saccade-related impairment.”

For elderly healthy adults:

“In contrast, colour memory remained unaffected by saccade demands ($F(3, 20) = 0.57$, $p = 0.65$, partial $\eta^2 = 0.03$), again supported by the Bayesian analysis: $BF_{01} = 8.68$, $P(\text{excl} | \text{data}) =$

0.90.”

For patient cohorts:

“Bayesian repeated measures ANOVAs further supported this dissociation, providing moderate evidence for the null hypothesis in the AD group ($BF_{01} = 3.35$, $P(\text{excl} | \text{data}) = 0.77$) and weak evidence in the PD group ($BF_{01} = 2.23$, $P(\text{excl} | \text{data}) = 0.69$). This indicates that even in populations with established neurodegeneration, the detrimental impact of eye movements is specific to the spatial domain.”

Related description is also updated in Methods – Statistical Analysis.

Minor:

(1) The modelling is described as computational but is arguably better characterised as a heuristic generative model at Marr's algorithmic level. It does not derive from normative computational principles or describe an implementation in neural circuits.

We appreciate your perspective on the classification of our model within Marr’s hierarchy. We agree that our framework is best characterised as an algorithmic-level generative model. Our objective was to identify the mechanistic principles governing transsaccadic updating rather than to provide a normative derivation or a specific circuit-level implementation.

To ensure readers do not over-interpret the term ‘computational’, we have added a clarifying statement in the Discussion acknowledging the algorithmic nature of the model. Interestingly, we note that a model predicated on this form of spatial diffusion implies a neural field representation with a spatial connectivity kernel whose limit approximates the second derivative of a Dirac delta function. While a formal neural field implementation is beyond the scope of the present work, our algorithmic results provide the necessary constraints for such future biophysical models.

p.20: “While we describe the present framework as ‘computational’, it is more precisely characterised as an algorithmic-level generative model within Marr’s hierarchy. Our focus was on defining the rules of spatial integration and the sources of eye-movement-induced noise, rather than deriving these processes from normative principles or defining their specific neural implementation.”

(2) I did not find a description of the recruitment and characterization of the AD and PD patients.

Apologies for this omission. We have now included a detailed description of participant recruitment and clinical characterisation in the Methods section and also updated Table 2:

“A total of 87 participants completed the study: 21 young healthy adults (YC), 21 older healthy adults (EC), 23 patients with Parkinson’s disease (PD), and 22 patients with Alzheimer’s disease (AD). Their demographic and clinical details are summarised in Table 2. Initially, 90 participants were recruited (22 YC, 21 EC, 25 PD, 22 AD); however, three individuals (1 YC and 2 PD) were excluded from all analyses due to technical issues during data acquisition.

All participants were recruited locally in Oxford, UK. None were professional artists, had a history of psychiatric illness, or were taking psychoactive medications (excluding standard dopamine replacement therapy for PD patients). Young participants were recruited via the University of Oxford Department of Experimental Psychology recruitment system. Older healthy volunteers (all >50 years of age) were recruited from the Oxford Dementia and Ageing Research (OxDARE) database.

Patients with PD were recruited from specialist clinics in Oxfordshire. All had a clinical diagnosis of idiopathic Parkinson’s disease and no history of other major neurological or

psychiatric conditions. While specific dosages of dopamine replacement therapy (e.g., levodopa equivalent doses) were not systematically recorded, all patients were tested while on their regular medication regimen ('ON' state).

Patients with PD were recruited from clinics in the Oxfordshire area. All had a clinical diagnosis of idiopathic Parkinson's disease and no history of other major neurological or psychiatric illnesses. While all patients were tested in their regular medication 'ON' state, the specific pharmacological profiles—including the exact types of medication (e.g., levodopa, dopamine agonists, or combinations) and dosages—were not systematically recorded. The disease duration and PD severity were also un-recorded for this study.

Patients with AD were recruited from the Cognitive Disorders Clinic at the John Radcliffe Hospital, Oxford, UK. All AD participants presented with a progressive, multidomain, predominantly amnesic cognitive impairment. Clinical diagnoses were supported by structural MRI and FDG-PET imaging consistent with a clinical diagnosis of AD dementia (e.g., temporo-parietal atrophy and hypometabolism).⁶⁹ All neuroimaging was reviewed independently by two senior neurologists (S.T. and M.H.).

Global cognitive function was assessed using the Addenbrooke's Cognitive Examination-III (ACE-III).⁷⁰ All healthy participants scored above the standard cut-off of 88, with the exception of one elderly participant who scored 85. In the PD group, two participants scored below the cut-off (85 and 79). In the AD group, six participants scored above 88; these individuals were included based on robust clinical and radiological evidence of AD pathology rather than their ACE-III score alone."

| (3) *YA and OA patients appear to differ in gender distribution.*

We acknowledge the difference in gender distribution between the young (71.4% female) and older adult (57.1% female) cohorts. However, we do not anticipate that gender influences the fundamental computational mechanisms of retinotopic maintenance or transsaccadic remapping. These processes represent low-level visuospatial functions for which there is no established evidence of gender-specific differences in precision or coordinate transformation. We have ensured that the gender distribution for each cohort is clearly listed in the demographics table (Table 2) for full transparency.

Thank you very much for very insightful feedback!

| **Reviewer #3 (Public review):**

Thank you for the positive feedback regarding our inclusion of clinical groups and the identification of computational phenotypes that differentiate these cohorts.

To address your concerns about the model, we have clarified our use of Bayesian Model Selection, which inherently penalises model complexity to ensure that our results are not driven solely by the number of parameters. We will also provide further evidence regarding model generalisability to address the concern of overfitting.

Regarding the link with the ROCF, we have revised the manuscript to better highlight the specific relationship between our transsaccadic parameters and the ROCF data and better motivate the inclusion of these results in the main text.

Below is our response to your suggestions point-by-point:

| (1) *The models tested differ in terms of the number of parameters. In general, a larger number of parameters leads to a better goodness of fit. It is not clear how the difference in the number of parameters between the models was taken into account. It is not clear whether the modelling results could be influenced by overfitting (it is not clear how well the model can generalize to new observations).*

To ensure our results were not driven by the number of parameters, we utilised random-effects Bayesian Model Selection (BMS) to adjudicate between our candidate models. Unlike maximum likelihood methods, BMS relies on the marginal likelihood (model evidence), which inherently balances model fit against parsimony—a principle known as the Occam's Razor (Rasmussen and Ghahramani, 2000). In this framework, a model is only preferred if the improvement in fit justifies the additional parameter space; redundant parameters actually lower model evidence by diluting the probability mass. We would be happy to point toward literature that discusses how these marginal likelihood approximations provide a more robust guard against overfitting than standard metrics like BIC or AIC (MacKay, 2003; Murray and Ghahramani, 2005; Penny, 2012).

The fact that the "Dual (Saccade) + Interference" model (Model 7) emerged as the winner—with a Bayes Factor of 6.11 against the next best alternative—demonstrates that its complexity was statistically justified by its superior account of the trial-by-trial data.

Furthermore, to address the risk of overfitting, we established the generalisability of these parameters by using them to predict performance on an independent clinical task. These parameters successfully explained ~62% of the variance in ROCF copy scores—a very distinct, real-world task—confirming that they represent robust computational phenotypes rather than idiosyncratic fits to the initial dataset.

In the Results (p10):

"We used random-effects Bayesian model selection to identify the most plausible generative model. This process relies on the marginal likelihood (model evidence), which inherently balances model fit against complexity—a principle often referred to as Occam's razor.[25–27]"

In the Discussion (p17):

"Importantly, the risk of overfitting is mitigated by the Bayesian Model Selection framework; by utilising the marginal likelihood for model comparison, the procedure inherently penalises excessive model complexity and promotes generalisability.[25–27,42] This generalisability was further evidenced by the model's ability to predict performance on the independent ROCF task, confirming that these parameters represent robust mechanistic phenotypes rather than idiosyncratic fits to the initial dataset."

(2) Results specificity: it is not clear how specific the modelling results are with respect to constructional ability (measured via the Rey-Osterrieth Complex Figure test). As with any cognitive test, performance can also be influenced by general, non-specific abilities that contribute broadly to test success.

We agree that constructional performance is influenced by both specific mechanistic constraints and general cognitive abilities. To isolate the unique contribution of transsaccadic updating, we therefore performed a partial correlation analysis across the entire sample. We examined the relationship between location error in the two-saccades condition (our primary behavioural measure of transsaccadic memory) and ROCF copy scores. Even after partialling out the effects of global cognitive status (ACE-III total score), age, and years of education, the correlation remained highly significant ($\rho = -0.39$, $p < 0.001$).

This suggests that our model captures a specific computational phenotype—the precision of spatial updating during active visual sampling—rather than acting as a proxy for non-specific cognitive decline. This mechanistic link explains why traditional working memory measures (e.g., digit span or Corsi blocks) frequently fail to predict drawing performance; unlike those tasks, figure copying requires thousands of saccades, making it uniquely sensitive to the precision of the dynamic remapping signals identified by our modelling framework.

We added the following text in the Discussion (p19):

“We also found that the relationship between transsaccadic working memory and ROCF performance remains highly significant ($\rho = -0.39$, $p < 0.001$), even after controlling for age, education, and global cognitive status (ACE-III total score). Consequently, transsaccadic updating may represent a discrete computational phenotype required for visuomotor control, rather than a non-specific proxy for global cognitive decline.[57]”

Reviewer #3 (Recommendations for the authors):

(1) The authors mention in the introduction the following: "One key hypothesis is that we use working memory across visual fixations to update perception dynamically", citing the following manuscript:

Harrison, W. J., Stead, I., Wallis, T. S. A., Bex, P. J. & Mattingley, J. B. A computational 906 account of transsaccadic attentional allocation based on visual gain fields. Proc. Natl. 907 Acad. Sci. U.S.A. 121, e2316608121 (2024).

However, the manuscript above does not refer explicitly to the involvement of working memory in transsaccadic integration of object location in space. Rather, it takes advantage of recent evidence showing how the true location of a visual object is represented in the activity of neurons in primary visual cortex (A. P. Morris, B. Krekelberg, A stable visual world in primate primary visual cortex. Curr. Biol. 29, 1471-1480.e6 (2019)). The model hypothesizes that true locations of objects are readily available, and then allocates attention in real-world coordinates, allowing efficient coordination of attention and saccadic eye movements.

Thank you for clarification. As suggested, we have now included the citation of Morris & Krekelberg (2019) to acknowledge the evidence for stable object locations within the primary visual cortex.

(2) The authors in the introduction and the title use the terms 'transsaccadic memory' and 'spatial working memory'. However, it is not clear whether these can be used interchangeably or are reflecting different constructs.

Classical measures of visuo-spatial working memory are derived from the Corsi task (or similar), where the location of multiple objects is displayed and subsequently remembered. In such tasks, eye movements and saccades are not generally considered, only memory performance, representing the visuo-spatial span.

Transsaccadic memory tasks are instead explicitly measuring the performance on remembered object locations of features across explicit eye movements, usually using a very limited number of objects (1 or 2, as is the case for the current manuscript).

While the two constructs share some features, it is not clear whether they represent the same underlying ability or not, especially because in transsaccadic tasks, participants are required to perform one or more saccades, thus representing a dual-task case.

I think the relationship between 'transsaccadic memory' and 'spatial working memory' should be clarified in the manuscript.

Thank you. Yes, we have added this within the Methods - Measurement of saccade cost to clarify that spatial working memory is the broad cognitive construct responsible for short-term maintenance, whereas transsaccadic memory is the specific, dynamic process of remapping representations to maintain stability across eye movements.

In Methods (p.22):

“Within this framework, it is important to distinguish between the broad construct of spatial working memory and the specific process of transsaccadic memory. While spatial working memory refers to the general ability to maintain spatial information over short intervals, transsaccadic memory describes the dynamic updating of these representations—termed remapping—to ensure stability across eye movements. Unlike classical 'static' measures of spatial working memory, such as the Corsi block task which focuses on memory span, transsaccadic memory tasks explicitly require the integration of stored visual information with motor signals from intervening saccades. Our paradigm treats transsaccadic updating as a core computational process within spatial working memory, where eye-centred representations are actively reconstructed based on noisy memories of the intervening saccade vectors.”

(3) In Figure 1, the second row indicates the presentation of item 2. Indeed, in the condition 'saccade-after-item-1', the target in the second row of Figure 1 is displaced, as expected. This clarifies the direction and amplitude of the first saccade requested. However, from Figure 1, it is hard to understand the amplitude and direction of the second requested saccade. I think the figure should be updated, giving a full description of the direction and amplitude of the second saccade as well ('saccade-after-item-2' and 'two-saccades' conditions).

We agree that making the figure legend more self-contained is beneficial for the reader. While the specific physical parameters and the trial sequence for each condition are detailed in the Results and Methods sections, we have now updated the legend for Figure 1 to explicitly define these details. Specifically, we have clarified that the colour wheel itself served as the target for the second instructed saccade (i.e., the movement from the second fixation cross to the colour wheel location). We have also included the quantitative constraint that all saccade vectors were at least 8.5 degrees of visual angle in amplitude. Given the limited space within a figure legend, we hope these concise additions provide the transparency requested without interrupting the conceptual flow of the diagram.

Updated Figure 1 legend:

“Participants were asked to fixate a white cross, wherever it appeared. They had to remember the colour and location of a sequence of two briefly presented coloured squares (Item 1 and 2), each appearing within a white square frame. They then fixated a colour wheel wherever it appeared on the screen, which served as the target for the second instructed saccade (i.e., a movement from the second fixation cross to the colour wheel location). This cued recall of a specific square (Item 1 or Item 2 labelled within the colour wheel). Participants selected the remembered colour on the colour wheel which led to a square of that colour appearing on the screen. They then dragged this square to its remembered location on the screen. Saccadic demands were manipulated by varying the locations of the second frame and the colour wheel, resulting in four conditions in their reliance on retinotopic versus transsaccadic memory: (1) No-Saccade condition providing a baseline measure of within-fixation precision as no eye movements were required. (2) Saccade After Item 1; (3) Saccade After Item 2; (4) Saccades after both items (Two Saccades condition). In all conditions requiring eye movements, saccade vectors were constrained to a minimum amplitude of 8.5° (degrees of visual angle). While the No-Saccade condition isolates retinotopic working memory, conditions (2) to (4) collectively quantify the impact of varying saccadic demands and timings on the maintenance of spatial information, thereby assessing the efficacy of the transsaccadic updating process.”

(4) The authors write: "Eye tracking analysis confirmed high compliance: participants correctly maintained fixation or executed saccades as instructed on the vast majority of trials (83% {plus minus} 14%). Non-compliant trials were excluded 136 from further analysis." 14% of excluded trials are a substantial fraction of trials, given the task

requirements. Is this proportion of excluded trials different between experimental groups, and are experimental groups contributing equally to this proportion?

We thank the reviewer for pointing this out, and we apologise for the confusion. The 83% trial number was actually across all four cohorts, and all conditions, and it was actually above 90% for YC, EC and even AD, but dropped to 60 ish in PD group.

We now have conducted a full analysis of compliant trial counts using a mixed ANOVA (4 saccade conditions x 4 cohorts). This analysis revealed a main effect of group ($F(3, 80) = 8.06$, $p < 0.001$), which was driven by lower compliance in the PD cohort (mean approx. 25.4 trials per condition) compared to the AD, EC, and YC cohorts (means ranging from 35.8 to 38.9 trials per condition). Crucially, however, the interaction between group and condition was not statistically significant ($p = 0.151$). This indicates that the relative impact of saccade demands on trial retention was consistent across all four groups.

Because our primary behavioural measure—the saccade cost—is a within-subject comparison of impairment across conditions, these differences in absolute trial numbers do not introduce a systematic bias into our findings. Furthermore, even with the higher attrition in the PD group, we retained a sufficient number of high-quality trials (minimum mean of ~23 trials in the most demanding condition) to support robust trial-by-trial parameter estimation and valid statistical inference. We have updated the Results and Methods to reflect these details.

In Results (p4):

“To mitigate potential confounds, we monitored eye position throughout the experiment. Eye-tracking analysis confirmed high compliance in healthy adults, who followed instructions on the vast majority of trials (Younger Adults: $97.2 \pm 5.2\%$; Older Adults: $91.3 \pm 20.4\%$). The mean difference between these groups was negligible, representing just 1.25 trials per condition, and was not statistically significant ($t(80) = 0.16$, $p = 1.000$; see more in Methods – Eyetracking data analysis). Non-compliant trials were excluded from all further analyses.”

In Methods (p27):

“Eye-tracking analysis confirmed high compliance overall, with participants correctly maintaining fixation or executing saccades on the vast majority of trials (83% across all participants). A mixed ANOVA revealed a main effect of group on trial retention ($F(3, 80) = 8.06$, $p < 0.001$, partial $\eta^2 = 0.23$), primarily due to lower compliance in the PD cohort (YC: $97 \pm 4\%$; EC: $91 \pm 10\%$; AD: $95 \pm 5\%$; PD: $63 \pm 38\%$). Importantly, there was no significant interaction between group and saccade condition ($F(3.36, 80) = 1.78$, $p = 0.15$, partial $\eta^2 = 0.008$), suggesting that trial attrition was not disproportionately affected by specific task demands in any group.

We acknowledge that this reduced trial count in the PD group represents a limitation for across-cohort comparison. However, the absolute number of compliant trials in PD group (mean approx. 25 per condition) remained sufficient for robust trial-by-trial parameter estimation. Furthermore, the lack of a significant group-by-condition interaction confirms that the results reported for this cohort remain valid and that our primary finding of a selective spatial memory deficit is robust to these differences in data retention.”

(5) Modelling

(a) Degrees of freedom, cross-validation, number of parameters.

I appreciate the effort in introducing and testing different models. Models of increase in complexity and are based on different assumptions about the main drivers and mechanisms underlying the dependent variable. The models differ in the number of

parameters. How are the differences in the number of parameters between models taken into account in the modelling analysis? Is there a cost associated with the extra parameters included in the more complex models?

(b) Cross-validation and overfitting.

Overfitting can occur when a model learns the training data but cannot generalize to novel datasets. Cross-validation is one approach that can be used to avoid overfitting. Was cross-validation (or other approaches) implemented in the fitting procedure against overfitting? Otherwise, the inference that can be derived from the modelled parameters can be limited.

To address your concerns regarding model complexity and overfitting, we would like to clarify our use of Bayesian Model Selection (BMS). Unlike frequentist methods that often rely on cross-validation to assess generalisability, we used random-effects BMS based on the marginal likelihood (model evidence). This approach inherently implements Bayesian Occam's Razor by integrating out the parameters. Under this framework, the use of the marginal likelihood for model selection provides a mathematically equivalent safeguard to frequentist cross-validation, as it evaluates the model's ability to generalise across the entire parameter space rather than just finding a maximum likelihood fit for the training data. Thus, models are penalised not just for the absolute number of parameters, but for their overall functional flexibility. A more complex model is only preferred if the improvement in model fit is substantial enough to outweigh this inherent penalty. The emergence of Model 7 as the winner (Bayes Factor = 6.11 against the next best alternative) confirms that its additional complexity is statistically justified.

Furthermore, in this study we provided an external validation of these recovered parameters by demonstrating that they explain 62% of the variance in an independent, real-world, clinical task (ROCF copy). This empirical evidence confirms that our model captures robust mechanistic phenotypes rather than idiosyncratic noise. We have updated the Results and Discussion to explicitly state these.

In Results: (p10)

"We used random-effects Bayesian model selection to identify the most plausible generative model. This process relies on the marginal likelihood (model evidence), which inherently balances model fit against complexity—a principle often referred to as Occam's razor.[26–28]"

In Discussion: (p17)

"Importantly, the risk of overfitting is mitigated by the Bayesian Model Selection framework; by utilising the marginal likelihood for model comparison, the procedure inherently penalises excessive model complexity and promotes generalisability.[26–28,43] This generalisability was further evidenced by the model's ability to predict performance on the independent ROCF task, confirming that these parameters represent robust mechanistic phenotypes rather than idiosyncratic fits to the initial dataset."

(6) n. of participants.

(a) The authors write the following: "A total of healthy volunteers (21 young adults, mean age = 24.1 years; 21 older adults, mean age = 72.4 years) participated in this study. Their demographics are shown in Table 1. All participants were recruited locally in Oxford." However, Table 1 reports the data from more than 80 participants, divided into 4 groups. Details about the PD and AD groups are missing. Please clarify.

We apologize for this lack of clarity in the text. We have rewritten and expanded the “Participants” section and corrected Table 2 in the Methods section to reflect the correct number of participants.

In Methods (p20):

“A total of 87 participants completed the study: 21 young healthy adults (YC), 21 older healthy adults (EC), 23 patients with Parkinson’s disease (PD), and 22 patients with Alzheimer’s disease (AD). Their demographic and clinical details are summarised in Table 2. Initially, 90 participants were recruited (22 YC, 21 EC, 25 PD, 22 AD); however, three individuals (1 YC and 2 PD) were excluded from all analyses due to technical issues during data acquisition.

All participants were recruited locally in Oxford, UK. None were professional artists, had a history of psychiatric illness, or were taking psychoactive medications (excluding standard dopamine replacement therapy for PD patients). Young participants were recruited via the University of Oxford Department of Experimental Psychology recruitment system. Older healthy volunteers (all >50 years of age) were recruited from the Oxford Dementia and Ageing Research (OxDARE) database.

Patients with PD were recruited from specialist clinics in Oxfordshire. All had a clinical diagnosis of idiopathic Parkinson’s disease and no history of other major neurological or psychiatric conditions. While specific dosages of dopamine replacement therapy (e.g., levodopa equivalent doses) were not systematically recorded, all patients were tested while on their regular medication regimen (‘ON’ state).

Patients with PD were recruited from clinics in the Oxfordshire area. All had a clinical diagnosis of idiopathic Parkinson’s disease and no history of other major neurological or psychiatric illnesses. While all patients were tested in their regular medication ‘ON’ state, the specific pharmacological profiles—including the exact types of medication (e.g., levodopa, dopamine agonists, or combinations) and dosages—were not systematically recorded. The disease duration and PD severity were also un-recorded for this study.

Patients with AD were recruited from the Cognitive Disorders Clinic at the John Radcliffe Hospital, Oxford, UK. All AD participants presented with a progressive, multidomain, predominantly amnesic cognitive impairment. Clinical diagnoses were supported by structural MRI and FDG-PET imaging consistent with a clinical diagnosis of AD dementia (e.g., temporo-parietal atrophy and hypometabolism).[70] All neuroimaging was reviewed independently by two senior neurologists (S.T. and M.H.).

Global cognitive function was assessed using the Addenbrooke’s Cognitive Examination-III (ACE-III).[71] All healthy participants scored above the standard cut-off of 88, with the exception of one elderly participant who scored 85. In the PD group, two participants scored below the cut-off (85 and 79). In the AD group, six participants scored above 88; these individuals were included based on robust clinical and radiological evidence of AD pathology rather than their ACE-III score alone.”

(b) As modelling results rely heavily on the quality of eye movements and eye traces, I believe it is necessary to report details about eye movement calibration quality and eye traces quality for the 4 experimental groups, as noisier data could be expected from naïve and possibly older participants, especially in case of clinical conditions. Potential differences in quality between groups should be discussed in light of the results obtained and whether these could contribute to the observed patterns.

Thank you for pointing this out. We have revised the Methods about how calibration was done:

(p27) “Prior to the experiment, a standard nine-point calibration and validation procedure was performed. Participants were instructed to fixate a small black circle with a white centre (0.5 degrees) as it appeared sequentially at nine points forming a 3 x 3 grid across the screen. Calibration was accepted only if the mean validation error was below 0.5 degrees and the maximum error at any single point was below 1.0 degree. If these criteria were not met, or if the experimenter noticed significant gaze drift between blocks, the calibration procedure was repeated. This calibration ensured high spatial accuracy across the entire display area, facilitating the precise monitoring of fixations on item frames and saccadic movements to the response colour wheel.”

Moreover, as detailed in our response to Point 4, while the PD group exhibited lower compliance, there was no interaction between group and saccade condition for compliance ($p = 0.151$). This confirms that any noise or trial attrition was distributed evenly across experimental conditions. Consequently, the observed “saccade cost” (the difference in error between conditions) is not an artefact of unequal noise but represents a genuine mechanistic impairment in spatial updating. We have updated the Methods to clarify this distinction.

Furthermore, our Bayesian framework explicitly estimates precision (random noise) as a distinct parameter from updating cost (saccade cost). This allows the model to partition the variance: even if a clinical group is “noisier” overall, this is captured by the precision parameter, ensuring it does not inflate the specific estimate of saccade-driven memory impairment.

(7) Figure 5. I suggest reporting these results using boxplots instead of barplots, as the former gives a better overview of the distributions.

We appreciate the suggestion to use boxplots to better illustrate data distributions. However, we have chosen to retain the current bar plot format due to the visual and statistical complexity of our 4 x 4 x 2 experimental design. Figure 5 represents 16 distinct distributions across four groups and four conditions for both location and colour measures; employing boxplots/violins for this density of data would significantly increase visual clutter and make the figure difficult to parse.

Furthermore, the primary objective of this figure is to reflect the statistical analysis and illustrate group differences in overall performance and highlight the specific finding that patients with AD were significantly more impaired across all conditions compared to YC, EC, and PD groups. Our statistical focus remains on the mean effects—specifically the significant main effect of group ($F(3, 318) = 59.71, p < 0.001$) and the critical null-interaction between group and condition ($p = 0.90$). The error measure most relevant to these comparisons is the standard error of the mean (SEM), rather than the interquartile range (IQR). We think that bar plots provide the most straightforward and scannable representation of these mean differences and the consistent pattern of decay across cohorts for the final manuscript layout.

To address the reviewer’s request for distributional transparency, we have provided a version of Figure 5 using grouped boxplots in the supplementary material (Supplementary figure 2). We note, however, that the spread of raw data points in these plots does not directly reflect the variance associated with our within-subject statistical comparisons.

(8) Results specificity, trans-saccadic integration and ROCF. The authors demonstrate that the derived model parameters account for a significant amount of variability in ROCF performance across the experimental groups tested (Figure 8A). However, it remains unclear how specific the modelling results are with respect to the ROCF.

The ROCF is generally interpreted as a measure of constructional ability. Nevertheless, as with any cognitive test, performance can also be influenced by more general, non-specific abilities that contribute broadly to test success. To more clearly link the specificity

between modelling results and constructional ability, it would be helpful to include a test measure for which the model parameters would not be expected to explain performance, for example, a verbal working memory task.

I am not necessarily suggesting that new data should be collected. However, I believe that the issue of specificity should be acknowledged and discussed as a potential limitation in the current context.

We appreciate this important point regarding the discriminant validity of our findings. We agree that cognitive performance in clinical populations is often influenced by a general "g-factor" or non-specific executive decline. However, we chose the ROCF Copy task specifically because it is a hallmark clinical measure of constructional ability that effectively serves as a real-world transsaccadic task, requiring participants to integrate spatial information across hundreds of saccades between the model figure and the drawing surface.

To address the reviewer's concern regarding specificity, we leveraged the fact that all participants completed the ACE-III, which includes a dedicated verbal memory component (the ACE Memory subscale). We conducted a partial correlation analysis and found that the relationship between transsaccadic working memory and ROCF copy performance remains highly significant ($\rho = -0.46$, $p < 0.001$), even after controlling for age, education, and the ACE-III Memory subscale score. This suggests that the link between transsaccadic updating and constructional ability is mechanistically specific rather than a byproduct of global cognitive impairment. We have substantially revised the Discussion to highlight this link and the supporting statistical evidence.

We first updated the last paragraph of Introduction:

"Finally, by linking these mechanistic parameters to a standard clinical measure of constructional ability (the Rey-Osterrieth Complex Figure task), we demonstrate that transsaccadic updating represents a core computational phenotype underpinning real-world visuospatial construction in both health and neurodegeneration."

The new section in Discussion highlighting the ROCF copy link:

"Importantly, our computational framework establishes a direct mechanistic link between transsaccadic updating and real-world constructional ability. Specifically, higher saccade and angular encoding errors contribute to poorer ROCF copy scores. By mapping these mechanistic estimates onto clinical scores, we found that the parameters derived from our winning model explain approximately 62% of the variance in constructional performance across groups. These findings suggest that the computational parameters identified in the LOCUS task represent core phenotypes of visuospatial ability, providing a mechanistic bridge between basic cognitive theory and clinical presentation.

This relationship provides novel insights into the cognitive processes underlying drawing, specifically highlighting the role of transsaccadic working memory. Previous research has primarily focused on the roles of fine motor control and eye-hand coordination in this skill.[4,50–55] This is partly because of consistent failure to find a strong relation between traditional memory measures and copying ability.[4,31] For instance, common measures of working memory, such as digit span and Corsi block tasks, do not directly predict ROCF copying performance.[31,56] Furthermore, in patients with constructional apraxia, these memory performance often remain relatively preserved despite significant drawing impairments.[56–58] In literature, this lack of association has often been attributed to "deictic" visual-sampling strategies, characterised by frequent eye movements that treat the environment as an external memory buffer, thereby minimising the need to maintain a detailed internal representation.[4,59] In a real-world copying task, the ROCF requires a high volume of saccades, making it uniquely sensitive to the precision of the dynamic remapping signals identified here. Recent eye-tracking evidence confirms that patients with AD exhibit

significantly more saccades and longer fixations during figure copying compared to controls, potentially as a compensatory response to transsaccadic working memory constraints.[56] This high-frequency sampling—averaging between 150 and 260 saccades for AD patients compared to approximately 100 for healthy controls—renders the task highly dependent on the precision of dynamic remapping signals.[56] We also found that the relationship between transsaccadic working memory and ROCF performance remains highly significant ($\rho = -0.46$, $p < 0.001$), even after controlling for age, education, and ACE-III Memory subscore. Consequently, transsaccadic updating may represent a discrete computational phenotype required for visuomotor control, rather than a non-specific proxy for global cognitive decline. [58]

In other words, even when visual information is readily available in the world, the act of drawing performance depends critically on working memory across saccades. This reveals a fundamental computational trade-off: while active sampling strategies (characterised with frequent eye-hand movements) effectively reduce the load on capacity-limited working memory, they simultaneously increase the demand for precise spatial updating across eye movements. By treating the external world as an "outside" memory buffer, the brain minimises the volume of information it must hold internally, but it becomes entirely dependent on the reliability with which that information is remapped after each eye movement. This perspective aligns with, rather contradicts, the traditional view of active sampling, which posits that individuals adapt their gaze and memory strategies based on specific task demands.[3,60] Furthermore, this perspective provides a mechanistic framework for understanding constructional apraxia; in these clinical populations, the impairment may not lie in a reduced memory "span," but rather in the cumulative noise introduced by the constant spatial remapping required during the copying process.[58,61]

Beyond constructional ability, these findings suggest that the primary evolutionary utility of high-resolution spatial remapping lies in the service of action rather than perception. While spatial remapping is often invoked to explain perceptual stability,[11–13,15] the necessity of high-resolution transsaccadic memory for basic visual perception is debated.[13,62–64] A prevailing view suggests that detailed internal models are unnecessary for perception, given the continuous availability of visual information in the external world.[13,44] Our findings support an alternative perspective, aligning with the proposal that high-resolution transsaccadic memory primarily serves action rather than perception.[13] This is consistent with the need for precise localisation in eye-hand coordination tasks such as pointing or grasping.[65] Even when unaware of intrasaccadic target displacements, individuals rapidly adjust their reaching movements, suggesting direct access of the motor system to remapping signals.[66] Further support comes from evidence that pointing to remembered locations is biased by changes in eye position,[67] and that remapping neurons reside within the dorsal "action" visual pathway, rather than the ventral "perception" visual pathway.[13,68,69] By demonstrating a strong link between transsaccadic working memory and drawing (a complex fine motor skill), our findings suggest that precise visual working memory across eye movements plays an important role in complex fine motor control."

We are deeply grateful to the reviewers for their meticulous reading of our manuscript and for the constructive feedback provided throughout this process. Your insights have significantly enhanced the clarity and rigour of our work.

In addition to the changes requested by the reviewers, we wish to acknowledge a reporting error identified during the revision process. In the original Results section, the repeated measures ANOVA statistics for YC included Greenhouse-Geisser corrections, and the between-subjects degrees of freedom were incorrectly reported as within-subjects residuals. Upon re-evaluation of the data, we confirmed that the assumption of sphericity was not violated; therefore, we have removed the unnecessary Greenhouse-Geisser corrections and corrected the degrees of freedom throughout the Results and Methods sections. We have ensured that

these statistical updates are reflected accurately in the revised manuscript and that they do not alter the significance or interpretation of any of our primary findings.

We hope that these revisions address all the concerns raised and provide a more robust account of our findings. We look forward to your further assessment of our work.

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